Quantum Physics For Absolute Beginners

Understand Quantum Physics and Its Real-World Applications Without Confusing Math

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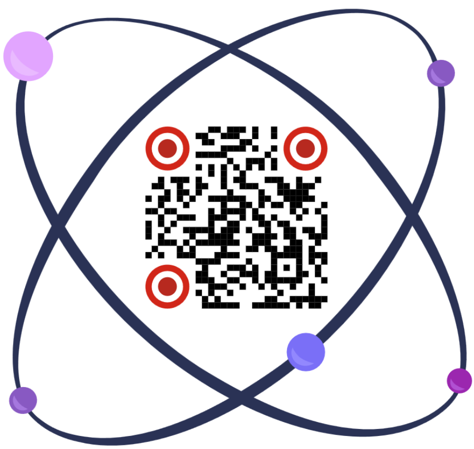

You will discover what are the milestones of Quantum Physics, from the early days to the latest discoveries, and the great minds that contributed to the Quantum Physics revolution.

If you want to who was the first Physicist to introduce the idea of the quantization of energy, make sure to download it now.

Introduction

I still remember my first encounter with Quantum Physics, as an undergrad student of Physics at the University of Florence. It was the Fall of 2004, more than 20 years ago, and I came out of that lesson, or better say, that semester of Quantum Physics lessons, quite shaken. At the end of each lecture, the blackboard was completely covered in formulas, which to be honest were not particularly difficult to understand. What drove me insane was that throughout the entire course, nobody mentioned what we were talking about. I mean, in a lecture about mechanics, you usually have a ball, an object of some kind, moving from one point to another, spinning around itself or whatever. During those lectures, we had matrices, vectors, differential equations, all the tools needed to solve quantum physics problems, but no context at all. We could have been talking for hours without mentioning any type of application. This is still the typical approach used in Universities around the world today. It's called axiomatic approach, where the first thing that is done is to present the rules of the game, and the rest of the time is devoted to an infinite amount of theorems and derivations, without any reference to why those axioms came to be formulated in the first place. It all eventually makes sense, but I had to find it out on my own and after months and maybe years of struggle and personal research.

This is probably one of the reasons why people struggle to understand Quantum Physics, even at a high level, without going into the math and theoretical foundation. It's not common for the people who went deep into the math to be able to take a step back, connect the experimental observations, all the historical facts and developments that eventually led to the modern formulation of quantum mechanics and its strange rules. I admit it's kind of useless for physicists to do so. After all, maybe a handful of physicists still alive have ever read the "Principia"[1] of Newton and there are better and easier ways to learn classical mechanics. But for Quantum Physics I feel like it's different, because the mathematical foundations and the principles on which the theory is based are completely at odds with our understanding of the world. It's not clear at all why those principles are needed, which are the facts that compelled us to think about the quantum world in new terms, and how the quantum world oddities can be reconciled with our

[1] The "Principia" Philosophiae Naturalis Principia Mathematica, is a foundational scientific book written by Sir Isaac Newton and first published in 1687. It is written in Latin and consists of three volumes that expound Newton's laws of motion and his universal law of gravitation.

everyday experience, governed by very different rules. That's why I think that to really understand Quantum Physics it's necessary to go through the same struggles that physicists at the beginning of the 20th century all went through. In other word, follow in their steps, look at the world with their eyes, understand their theories and experiments and eventual breakthroughs. This is the approach that I take in this book.

We will start our journey with a review of classical physics, which is what was known to physicists at the beginning of the quantum revolution. Classical physics, mechanics, acoustics, optics, electromagnetism, etc. studies phenomena that are directly accessible to us, meaning phenomena that we can see, hear, or touch. Concepts such as force and motion are quite familiar to us and our way of viewing nature, even if we do not know how to formulate Newton's equations. Classical physics attempts to provide a quantitative and mathematical description of what we observe, and this is done through the formulation of physical laws, which we then try to extrapolate from everyday observations to other situations. For example, the motion of billiard balls is well described by Newton's laws. Extending these laws to the astronomical scale, the motion of planets is also accurately described by the same laws. This reinforces the idea that Newtonian mechanics can apply to objects of any size. Therefore, in the second half of the last century, when Maxwell, Boltzmann, and others began studying atomic and molecular properties, they naturally assumed that Newton's laws and electromagnetism still provided a good description of these phenomena. For example, a gas was thought of consisting of very small atoms that interacted with each other exactly like small billiard balls, and these interactions were ultimately responsible for the macroscopic properties of gases, like temperature, pressure and so on.

Electrons were also thought of as very small particles, with definite positions and velocities at any instant. However, we will see how experiments showed that this assumption is incorrect in many ways, and how many of the properties that apply to our macroscopic world, completely break down when applied to the atomic scale. The problem is that the behavior of matter at this scale does not correspond at all to our everyday experience. For example, if I tell you that there is a way to go around a tree other than passing to its left or right, you would look at me like I was a madman. But for an electron, experiments show that yes, an electron can pass simultaneously on both sides, just like a beam of light that splits when encountering an obstacle. This phenomenon, which seems absurd to our intuition, has been experimentally confirmed in many ways.

During our journey into quantum physics we will focus with particular attention to the various experiments that either challenged the traditional view of physics, or confirmed the new quantum theories. This will help us understand how quantum physics principles have been repeatedly tested and verified, even if they are not accessible to our everyday experience and require physicists to probe the microscopic scale of atoms and the constituents of matter, where quantum properties manifest themselves.

Despite being formulated more than 100 years ago, Quantum Physics is more and more relevant today. The technologies related to quantum physics are getting more and more sophisticated, and the importance of Quantum Physics in the advancement of human knowledge is highlighted also by the Nobel Prize in Physics awarded recently. The 2022 Nobel Prize recognized the pioneering experiments related to entanglement, a typical Quantum Physics phenomenon. More recently, the 2025 Nobel Prize was awarded for groundbreaking work on quantum behavior in macroscopic electrical circuits, involving the observation on a macroscopic scale of another property of objects following the rules of Quantum Physics, the quantum tunneling.

I wrote this book for lifelong learners and curious minds who seek to understand the fascinating world of quantum physics without being overwhelmed by complex mathematics. It is for those who want to grasp the core concepts and fundamental issues that define quantum mechanics, providing a clear and accessible path into this fundamental topic. Whether you are a student, a professional from another field, or simply a curious reader, this book offers explanations that focus on intuition and insight, without unnecessarily heavy formalism often encountered in traditional textbooks. The goal is to make quantum physics approachable and engaging, to spark curiosity while at the same time delivering a rigorous explanation. I hope that this introduction to quantum physics will make you appreciate the strange and exciting behavior of particles and waves at the smallest scales, opening the door to further exploration with the help of more advanced material. Of course, some of the topics treated in the book will be simplified, but if you are a reader that requires more details or a deeper dive on certain aspects, from historical developments to experimental details, I have put additional references at the end of each chapter. Those are either technical articles, or books, or divulgation articles on specialized magazines, often written by the very fathers of the field. I am sure you will appreciate them.

Enough with the introduction. Let's dive in.

1 Physics at the dawn of the 20th century

At the dawn of the 20th century, physics seemed a well established discipline. Most physicists were busy making discoveries around the applications of classical mechanics, while electromagnetism and thermodynamics gradually gained significance. A few challenges arose at the intersections of these fields, notably in understanding heat radiation (the emission of light by objects warmed up at high temperature) and heat capacity (how good are objects at absorbing heat), for which definitive explanations remained elusive.

Today we call these fields Classical Physics. Of course at the time it was the only type of Physics, but today we know that these theories are only approximations and fail at very small scales. Even though they still describe pretty well most of the phenomena that we interact with every day (what we usually call the macroscopic scale). Before diving into Quantum Physics, which explains the world at microscopic scales as well as macroscopic one, it's important to understand the state of classical physics before the quantum revolution.

1.1 Mechanics

Mechanics studies the motion of particles and systems of particles, as well as the forces that cause or affect this motion. It is divided into three main areas.

- Kinematics, describes motion without regard to forces.
- Dynamics, studies forces and their effects on motion
- Statics, examines bodies at rest.

The origin of classical mechanics is considered to be the release of Isaac Newton's laws of motion alongside his pioneering development of calculus in 1687. Newton's 3 laws of motion could be formulated as follows

1. A body stays at rest, or it moves with a constant speed in a straight line, unless a force acts upon it.
2. At any instant, the force on a body is equal to the body's acceleration multiplied by its mass.
3. When two bodies exert forces on each other, the forces must have the same magnitude and opposite direction.

The application of these laws helped explain various physical phenomena, from the movement of planets to the analysis of forces acting on structures to the prediction of ballistics trajectories.

Over the following two centuries, the analytical methods of mechanics advanced significantly, and the concept of energy was developed. Kinetic energy was defined as the energy an object possesses due to its motion, and it depends on the object's mass and velocity. Potential energy, on the other hand, is the energy stored in an object due to its position or condition. It represents the potential to do work based on the object's placement in a force field (such as gravitational) or its state. Common examples include the energy stored in an elevated object due to gravity or in a compressed spring. Potential energy can be converted to kinetic energy as the object changes its position or state.

Over the decades, some alternative useful mathematical formulations of mechanics were introduced by Hamilton and Lagrange, who also introduced the principle of stationary action. According to this principle, the actual path taken by a physical system between two states is the one for which the action is stationary (usually a minimum). The action S is defined as the sum over time of the Lagrangian (kinetic energy minus potential energy) of the system. In essence, among all possible paths connecting initial and final states, the true path makes the action neither increase nor decrease for small changes around that path. This principle is fundamental in deriving the equations of motion using calculus of variations, and it underlies much of classical and quantum mechanics.

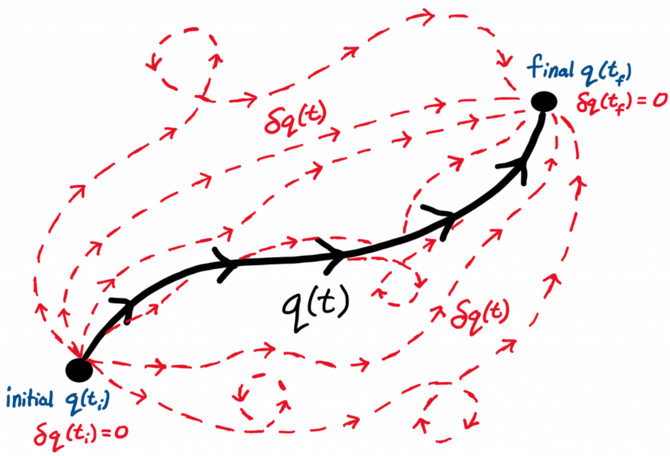

Figure 1.1: *Classically, a particle only has one true path that it follows between two points (illustrated in black). However, there are an infinite amount of surrounding paths with the same initial and final destinations that the particle could have taken instead (illustrated in red). Among all the red paths, the true path is the one that causes the action to be stationary, thus obeying the laws of physics. Source: saphysics.com*

Mechanics also helped to model the phenomenon of waves, which are disturbances that transfer energy from one place to another through a medium or space without the permanent displacement of the medium itself. In other words, they are propagation of energy that doesn't involve permanent movement of particles. Mechanical waves can propagate through air, water, or solids. For example, sound is a mechanical wave that propagates through air. When we clap our hand we compress the molecules of air between our hands, these molecules move very fast and hit the others in their vicinity, which in turn hit the nearby ones and so on. This causes a cascade effect where the kinetic energy moves from a group of air molecules to the closest ones and propagates without those molecules permanently moving, but only transferring energy to each other. These energy vibrations in air is what our ears perceive as sound.

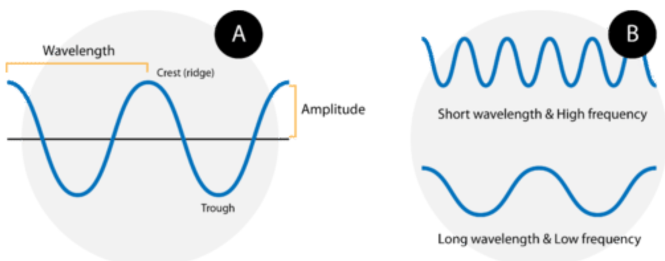

Figure 1.2: *A wave consists of alternating crests and troughs. The wavelength is the distance between any two consecutive identical points on the waveform. A short-wavelength wave (top) has a high frequency since more waves pass a point in a definite amount of time. Following the same logic, a wave with a longer wavelength (bottom) has a lower frequency. Source Christopher Auyeung via CK-12 Foundation.*

Mechanical waves can be classified mainly into three types:

- Transverse waves, where particles of the medium move perpendicular to the wave direction (water waves).

- Longitudinal waves, where particles move parallel to the wave's direction, creating compressions and rarefactions (sound waves).
- Surface waves, which have characteristics of both transverse and longitudinal waves, occurring at interfaces such as water surfaces.

I will now define two concepts that are extremely important in Quantum Physics, and I will mention them many times in the book. Frequency and wavelength are the fundamental properties of waves that describe their behavior and characteristics. Frequency is the number of complete wave cycles that pass a given point per second. It is measured in Hertz (Hz) and represents how often the wave oscillates in time. Higher frequency means more cycles per second, corresponding to higher energy and, for sound, a higher pitch, or for light, a shift towards blue colors. Wavelength is the physical distance between two consecutive points that are in phase on the wave, such as crest to crest or trough to trough. It is usually denoted by the Greek letter λ (lambda) and measured in meters (or centimeters, nanometers depending on context). Longer wavelengths translate into lower frequencies, while shorter wavelengths translate into higher frequencies.

1.2 Optics

Newton's work on mechanics proved to be useful to study not only the movement of bodies, but also the trajectory of light and its properties, advancing significantly the field of optics. Optics is the branch of physics that studies light, how it generates, propagates, and interacts with matter. It is divided into geometrical optics, which treats light as rays and analyzes reflection and refraction, and physical optics, which treats light as waves and covers phenomena such as diffraction and interference. These two phenomena are very important to understand the nature of light, and it's important to define them clearly, so we can understand the role they play in Quantum Physics.

Diffraction happens when waves spread out, bend around obstacles or pass through narrow openings, and it's the phenomenon responsible for waves changing direction and forming new patterns. This effect is most noticeable when the size of the obstacle or opening is comparable to the wavelength of the waves.

Refraction is instead the change in direction of a wave as it passes from one medium into another with a different density or refractive index, for example from water to air. Any wave changes speed when passing from medium to another, and this change in speed causes the wave to bend at the boundary between the two media. Common examples include light bending when passing from air into water or sound waves changing speed as they move between different materials. Refraction is responsible for phenomena like lenses focusing light and the apparent bending of objects submerged in water.

Polarization is another important property, typical of transverse waves. It describes the geometrical orientation of their oscillations relative to the direction of the wave. In electromagnetic waves like light, polarization refers specifically to the direction in which the electric field vibrates perpendicular to the wave's direction of propagation. Light waves can be linearly polarized, where the electric field oscillates consistently in a single plane. But they also can be circularly or elliptically polarized. In these cases, the electric field rotates around the direction of travel.

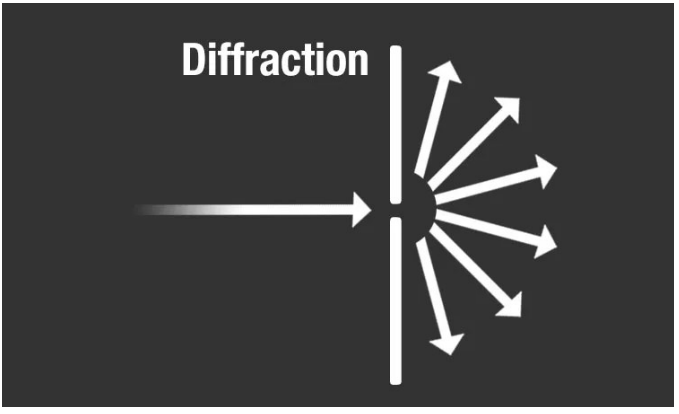

Figure 1.3: *Diffraction is the bending and spreading of waves around an obstacle. Source: NASA.*

1.3 Thermodynamics

Thermodynamics was probably the most popular field in Physics throughout the 19th century, and giants like Maxwell and Planck contributed greatly to the development of the field. In short, thermodynamics studies the relationships between heat, work, temperature, and energy. It deals with how energy transfers from one form to another and how these transfers affect the physical properties of matter.

Historically, thermodynamics developed contextually to the invention and development of the early steam engines, and its objective was to understand and improve the efficiency of those machines. After those initial efforts, the applications of thermodynamics where quickly extended to chemical compounds and chemical reactions. Of great importance for the development of quantum theory was the statistical approach to thermodynamics, also known as statistical mechanics. With the development of atomic theories in the late 19th century, physicists efforts focused towards correlating the microscopic properties of individual atoms and molecules to the macroscopic effects observed in thermodynamics. The application of Newtonian mechanics to many atoms, coupled with statistical considerations on the properties of those atoms, like their kinetic energy, allowed physicists to explain classical thermodynamics as a natural result of the behavior of the underlying particles that matter is made up of.

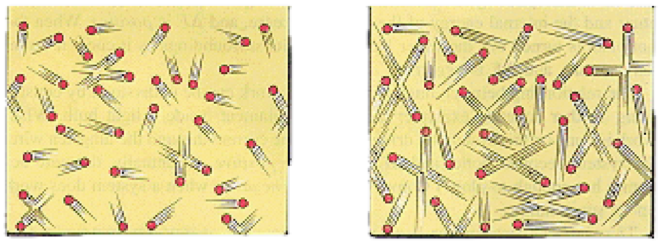

Figure 1.4: *Temperature is a measure of the average kinetic energy of the particles (atoms or molecules) in a substance. As temperature increases, particles move faster, with higher velocity and kinetic energy.*

1.4 Electromagnetism

Electrical and magnetic phenomena were known to physicists since Ancient Greece, but it took several centuries to finally develop a coherent theory that describes those forces, that are the expression of a single electromagnetic force. Electromagnetism deals with the interaction between electrically charged particles through electric and magnetic fields. It is one of the four fundamental forces of nature and combines electricity and magnetism into a single unified force. This interaction is responsible for many daily phenomena, such as the attraction and repulsion of charged particles or the binding of atoms into molecules.

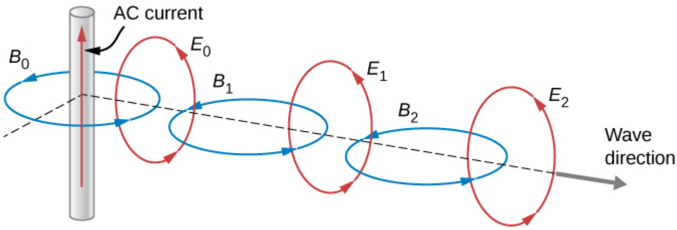

Figure 1.5: *An electromagnetic wave consists of two perpendicular waveforms (one for the electric and one for the magnetic field) in phase with one another. Source technologyuk.net*

At its core, electromagnetism describes how electric charges produce electric fields and how moving charges generate magnetic fields. These fields exert forces on other charges and moving particles, which can be observed as electric forces or magnetic forces depending on the situation. The theory of electromagnetism is mathematically described by Maxwell's equations, which also explains how electric and magnetic fields propagate as electromagnetic waves, including visible light, radio waves, and X-rays. These equations were formulated in 1860, and they showed that oscillating electric and magnetic fields can sustain each other and propagate through space as waves without needing a material medium. Maxwell mathematically derived that these waves travel at a fixed speed in vacuum, which matches the experimentally known speed of light, leading to the revolutionary conclusion that light itself is an electromagnetic wave.

1.5 Clouds on the horizon

In 1894, physics seemed complete. The theory of electromagnetism perfectly explained electrical and magnetic phenomena, and even predicted the nature of light as an electromagnetic wave, together with its speed. But not everybody in the Physics world was convinced. Lord Kelvin spotted two looming “clouds” on the horizon, and on 27 April 1900 he gave a lecture to the Royal Institution, entitled “Nineteenth-Century Clouds over the Dynamical Theory of Heat and Light”. The two “dark clouds” he talked about were:

- The question of the existence and the nature of the luminiferous ether, especially in light of the troubling null results of the Michelson-Morley experiment, which cast doubt on the ether’s motion relative to matter.
- The failure of the Maxwell-Boltzmann equipartition theorem to explain certain experimental findings, notably specific heat measurements in gases that did not conform to classical expectations.

The first issue relates to the fact that electromagnetic waves propagated in vacuum, without the need of a medium, which was a very strange phenomenon. Moreover, those waves seemed to move at a constant speed (the speed of light) even when the object emitting the wave, or the apparatus detecting them was moving at some speed towards them. In other words, this speed seemed to be a constant of nature, independent of the reference frame. We will not go into much detail on how this issue was solved, but you might have already guessed that Einstein solved it with the development of the theory of special (and later general) relativity.

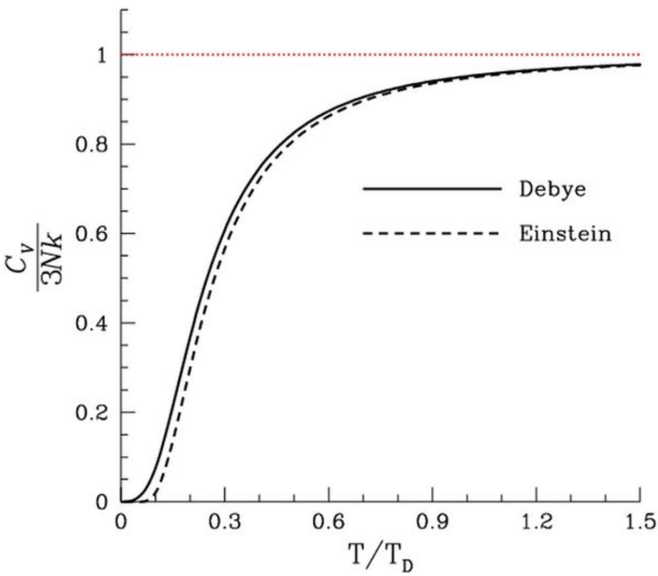

Figure 1.6: *Heat capacity as a function of temperature, according to Einstein's and Debye's later model. Heat capacity is zero at absolute zero, and rises to a constant value as the temperature becomes high. The red line corresponds to the classical limit of the Dulong–Petit law. Source Wikipedia*

The second issue was related to strange effects and experimental observations that did not agree with the statistical theory of thermodynamics, specifically with the heat capacity of solids. The heat capacity represents how much heat is required to increase the temperature of a solid, and it helps in understanding how solids absorb and store thermal energy. In solids, atoms cannot translate (unlike those in gases), but are free to vibrate about their equilibrium positions. Now, if you analyze such a system according to the classical theory of vibrations, you find out that the heat capacity of solids is constant and doesn't change as the temperature change. The solid should absorb the same heat when going from 0 to 1 degrees Celsius or going from 1000 to 1001 degrees Celsius. This is also called the Dulong-Petit law. The problem is, this result is in blatant disagreement with the third law of thermodynamics.

What does the third law state? Ok, let's take this step by step. Firstly, let's define entropy. Entropy is a measure of the degree of disorder, randomness, or the number of possible microscopic arrangements of atoms and molecules in a system. In the case of a crystal, it can be calculated simply by counting the number of ways atoms can vibrate around their equilibrium positions. The third law states that the entropy of a crystal should approach zero as the temperature drops to zero. In other words the number of ways atoms can vibrate should decrease with the temperature. And since entropy is related to heat capacity, this too should approach zero as the temperature increases.

So, the fact that classical theory of vibrations led to the conclusion that the heat capacity of solids does not depend on the temperature, seemed to be a failure of statistical physics, which was otherwise quite successful in predicting the macroscopic properties of matter, starting from the microscopic movements of its constituents. This disagreement was solved by Einstein (again) who explained the way atoms can vibrate in solids, by introducing the hypothesis of quantization of energy.

Believe it or not, Einstein was not the first to come up with this idea. The first person to introduce the revolutionary quantum hypothesis was Planck, in his famous article on the black body spectrum. We will dive into it in the next chapter.

1.6 Physics at the dawn of the 20th century: 3 takeaways

At the dawn of the 20th century, Physics seemed a well established discipline. Mechanics explained fairly well the static properties and movement of objects like planets. Thermodynamics was understood as the macroscopical manifestations of an immense amount of microscopic atoms interacting with each other according to classical mechanics. Electrical and magnetic phenomena were explained according to Maxwell's equations, which also predicted light as an electromagnetic wave. Today we refer to these theories as classical physics.

Despite the large success of classical physics, there were still unexplained observations and challenges. Lord Kelvin talked about two "clouds" on the horizon, the question of the existence of ether, through which electromagnetic waves were supposed to move, and the disagreement between the heat capacity of solids with the third principle of thermodynamics.

Both clouds were dissipated by Einstein. The existence of ether and the related issue of the constant speed of light was solved by the theory of relativity. The problem of the heat capacity of solid was solved with the help of the quantum hypothesis, firstly introduced by Planck.

1.7 To know more

- A fairly deep dive into Newton's laws directly from NASA
  - https://tinyurl.com/ms6vfr79
- For a more deep introduction to the various subjects that make up classical physics I warmly recommend Susskind's series "The theoretical minimum". You can buy his books everywhere now, but maybe you didn't know that you can also find all his courses on the website.
  - https://theoreticalminimum.com
- If you are interested in the history of the formulation of the principle of least action, one of the fundamental blocks of classical physics and beyond, I find this article a very nice historical overview of the subject.

    - https://tinyurl.com/56vmn3jc

- The famous speech by Lord Kelvin has often been misinterpreted. This article helps to clarify all issues he raised in his famous speech.
    - https://arxiv.org/abs/2106.16033

- Classical physics is just an approximation of quantum physics on the macroscopic scale. To understand exactly how this happens and what it exactly means, this article explains it perfectly
    - https://tinyurl.com/yktmuy2b

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2 The Black Body Problem

At the turn of the 20th century, a 43-year-old man, born in Kiel, Germany, had the idea that changed Physics forever. In an almost desperate attempt to justify the experimental observations of how the color of an object changes, as its temperature increases, he came up with the intuition that energy is not emitted continuously, but in small discrete "quanta", solving the problem of the black-body and giving birth to the Quantum Physics era. But to understand the significance of Planck's discovery and how it set the stage for a revolution in Physics, we need to go back to 1859.

2.1 Temperature and light

Every object emits light, but most of the objects that surround us emits infrared light. This is light that has a frequency too low to be able to be seen with our eyes. As the temperature of an object increases, it starts to emit more and more light at higher frequencies, that is visible to us (in the visible spectrum). You are surely familiar with the incandescence effect on bodies, where very hot materials start to glow. For example, iron heated to very high temperatures by blacksmiths, red burners on electric stoves, hot lava flowing from a volcano or the Sun.

Every object that is heated to high enough temperature, starts to emit visible light, first red, at around 480 C (900 F) all the way to orange, yellow and white at 1300 C (2400 F). The Sun surface is approximately at 5500 C (9980 F), and at that temperature, an object emits light at all frequencies in the visible spectrum, looking as a white color to our eyes.

Physicists were investigating light since Newton, and in the 18th century they were busy trying to understand the relationship between the light emitted by hot bodies, which back then was called heat rays, and visible light rays coming from a light source like the Sun. It was Gustav Kirchhoff that discovered in the mid-18th century, that there really is no difference between these heat rays and light rays, and proposed to study them by creating a so called black-body in laboratory. But what exactly is a black-body?

2.2 The black-body experiment

A perfect black-body is an ideal object that does not exist in nature. Physicists often use these types of concepts to clarify an idea, by making simple assumptions and derive a model that can be applied and tested in a real world experiment. This ideal object has the interesting property to completely absorb all light that falls on it, no matter the color of light (which depends on its wavelength, or frequency). This is why it's called black. And yet, even if it absorbs all light that falls on it without reflecting any of it, this body still emits light, because of its temperature.

A perfect black body does not exist in nature. However, Kirchhoff again had a good idea on how to build one. If one could build a large box with a small hole on the side, this will be an excellent absorber. Any radiation that goes through the hole bounces around inside, a lot of it getting absorbed on each bounce, and has little chance of ever getting out again. In reverse, if one could build an oven with a tiny hole on the side, the radiation coming out the hole would be a very good approximation of a perfect emitter.

A black-body of this type was first built and used for measurements in the 1890s in Berlin. Its design has been used with very few changes to this day, and it consists in making a small hole in a platinum box, with its interior blackened by iron oxide. The cavity is then heated up to a specific

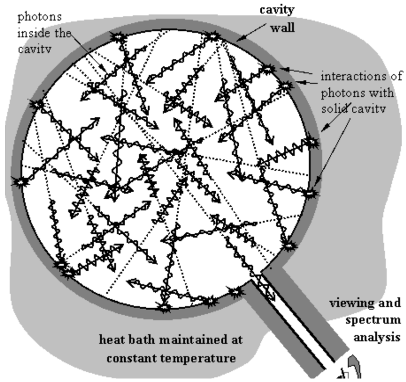

Figure 2.1: *Diagram of an ideal black-body. The light is represented as photons (more on this later) that interact with the internal walls by being absorbed or reflected. Only a small fraction of light escapes the black-body and can be detected. Source: University of Toronto Physics Dpt.*

temperature. The light coming from the hole is then analyzed with the help of a spectrometer[1], which measures the intensity and frequency of the radiation emitted from the cavity. The last step is to measure the intensity of these colors, which in the original 1890 experiment was done with the

[1] A simple spectrometer can be realized with the help of a prism. When a light beam made of multiple frequencies enters the prism, its direction changes slightly. However, the change of direction is not the same for all the frequencies that make up the light beam. So what happens is that red light is deviated in one direction, green light is slightly more deviated, blue a bit and more and so on. In the end you obtain a separation of the initial light beam into different colors, which are its frequency components.

help of a bolometer. This was a made of a thin metal strip, which electrical resistance changed with temperature. The beams impacting the strip caused a slight change in its resistance, which was measured as an electrical signal (for example a voltage change) in a circuit.

By measuring this signal at different frequencies physicists were able to obtain a curve of light intensity as a function of the light frequency emitted by a black-body. In other words, they measured the black-body spectrum. The measurements showed that the spectrum started from 0 at very low frequencies, had a peak at some frequency, and then again decreased to 0 at very high frequencies. And by increasing the temperature of the cavity, they were able to observe how the peak of the radiation shifted from lower to higher frequencies. In other words, the warmer the black-body, the more light emitted towards the blue end of the spectrum. Soon enough, in 1896, Wien found a radiation law that was in convincing agreement with the precise measurements performed at the laboratories. The problem was, the Wien formula had no theoretical foundation. In other words, it worked, but nobody could say why.

2.3 Oscillators might be the answer

Many physicists at the time were dissatisfied with the Wien law. Surely, it fit reasonably well the experimental data, but completely failed to explain why. It was just a practical law, that gave physicists no insights into how a black-body system really worked. Many of them started to look for a more fundamental law, one that could arise from applying statistical considerations to fundamental physics, like the radiation emitted by an oscillator.

Oscillators in Physics are a very practical idea and are a very good approximation of numerous physical systems. In practice, an oscillator is a system that goes under periodic motion around a stable equilibrium point. The motion is called oscillation, and it involves the system moving back and forth or rotating around an axis. The pendulum is a good example of an oscillator. A simple pendulum is made up of a weight suspended from a support. The equilibrium point of the system occurs when the weight is perfectly vertical. Once you move the weight, it starts to swing under gravity, with an oscillation that is periodic, meaning it always takes the same time for the pendulum to swing from one side to the other.

In early 1900, applying the same principle, Rayleigh had built a model of the black body

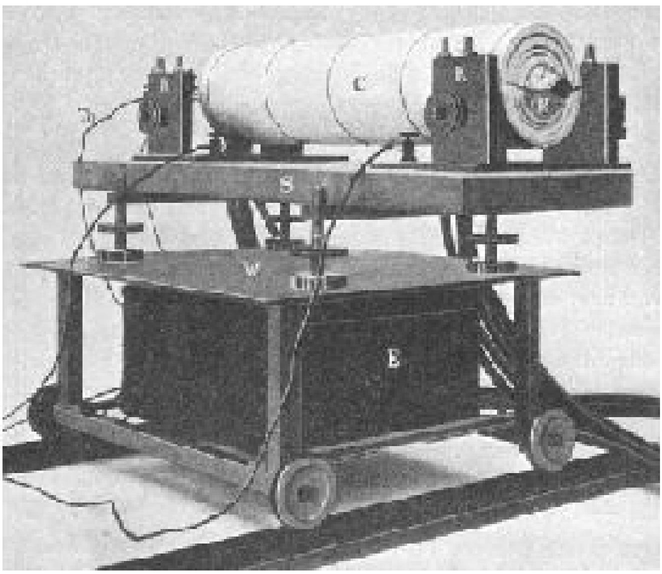

Figure 2.2: *The black body cavity used for the original experiment in 1890. The cylinder is a hollow cavity with blackened inside walls. Source: Phys. Bl. 12/2000, p. 43*

as a multitude of electrically charged oscillators. Back then physicists didn't have a clear idea of what matter was made of, but they believed that there existed a unit of matter, called atom, that contained both positively charged and negatively charged particles. The idea was that, due to temperature, those atoms movements made the atomic electric charges oscillate around an equilibrium point. And Maxwell equations predicted that an electrically charged particle oscillating around a point, emits energy in the form of light, which explains why a heated body emits light. Rayleigh's idea was that, as the temperature gets higher in the black-body, the movement of the atoms inside becomes faster, making it possible for faster oscillations to occur, which in return leads to more light emitted at higher frequencies. So we end up in a situation where some oscillators have low energy, some have high energy, but as the temperature gets higher, a larger fraction of them will have a high energy and thus emit light at high frequencies.

The so-called Rayleigh-Jeans law proved to be quite accurate at low frequencies, but it predicted that the intensity of the radiation emitted would increase with the frequency indefinitely. In other words, it predicted that, for any body at any temperature, you should see more light emitted in the blue-violet range than in the red-orange one. And even more light in the ultraviolet region and so on indefinitely. This is clearly in complete disagreement with the experimental results and more importantly goes against common sense. In fact, it leads to the absurd conclusion that there is an infinite amount of energy inside a black-body cavity.

This is the famous “ultraviolet catastrophe”, which is a direct conclusion of applying classical physics principles to model the radiation of the black-body. It's the failure of classical physics and the first signal to the physicist of the early 20th century, that everything they knew until then was based on completely wrong assumptions, and that new principles had to be introduced in order to formulate new laws of nature. It's quite interesting how a macroscopic effect like the observation of light coming from a hot body, could give us such a strong signal that something was very wrong with our understanding of the World.

2.4 Planck's solution

It was Planck who had the right intuition and introduced the first radical change in the theory. He was working on finding an explanation to the empirical Wien law, “at any cost, no matter how high”. And he started in the same way as Rayleigh, by modeling the black-body as a system made of many oscillators.

From the behavior of those microscopic oscillators he then derived the macroscopic behavior of the black-body, and calculated how much light

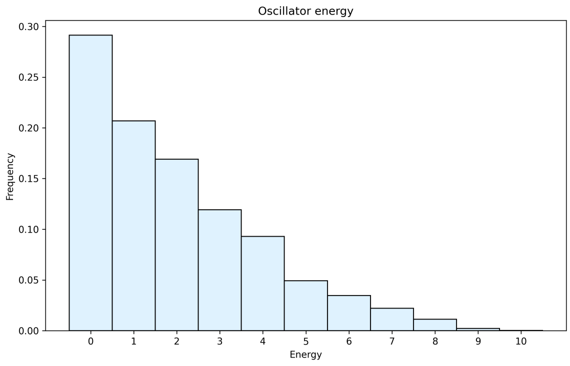

Figure 2.3: *A simulation of the way the energies of oscillators are likely to be distributed in a system with a total amount of energy fixed at 10.*

is produced by the whole system of oscillators at different frequencies. To do this, he divided the total energy in the system in small intervals, and started to calculate the number of ways the total energy could be distributed in those intervals, over a number of oscillators. In other words, he wanted to count how many configurations of microscopic oscillator energies lead to the same macroscopic black-body energy.

To illustrate the point, let's assume the total energy of the black-body is 10. We will divide the intervals from 0-1, 1-2 and so on, and we have 5 oscillators in the cavity. The table below shows one of the possible configurations.

Energy	0	1	2	3	4	5	6	7	8	9	10
Oscillators	1	1	2	0	0	1	0	0	0	0	0

Here we have:

- 1 oscillator with energy 0
- 1 oscillator with energy 1, total energy is 1
- 2 oscillators with energy 2, total energy is 4
- 1 oscillator with energy 5, total energy is 5

If we sum up the energies we find the total energy which is 10. There are many other combinations that can lead to the same amount of energy in the system, and the most probable configuration will also be the one that will happen in nature. So we can see here that there is only one configuration where there is only one oscillator that takes all the energy, and the rest have zero. While there are many configurations where each oscillator is in a low energy state, with few having higher energy. The most probable distribution will look something like the picture below.

Planck's final step was to assume that the total energy of the black-body is divided into finite portions of energy, not just as a mathematical trick, but as a reality of nature. This is the principle of quantization of energy, and Planck was the first to use it, to explain a physical fact. Planck himself called this an "act of desperation".

This shows how incredibly absurd and counter-intuitive this idea sounded at the time. Planck

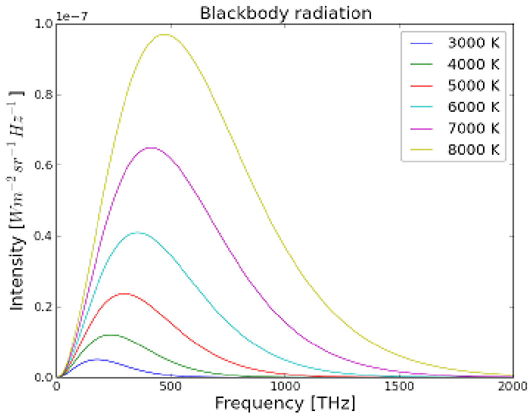

Figure 2.4: *The spectrum of the black body as predicted by Planck's law.*

was forced to introduce the quantization of energy, simply because nothing else would work, and no other approach would be able to explain the experimental observations. By introducing the hypothesis that the energy of each oscillator is equal to the frequency of the radiation that it emits, multiplied by a constant h (the Planck's constant), it follows that the higher the frequency of the light emitted by an oscillator, the higher its energy.

As we have seen, the black-body system is in a configuration where there are very few oscillators that have a high energy. And since, according to Planck's hypothesis, the energy of an oscillator is proportional to the frequency of the light emitted, it follows that there will be very few oscillators that emit light at high frequency. This was in agreement with the observations from the experiments at the time and provides a coherent explanation of the black-body spectrum, without any ultraviolet catastrophe, perfectly explaining why the intensity of the black-body radiation at very high frequency is very low.

2.5 The birth of quantum physics?

History tells that Planck himself was not aware of the impact of his work when he presented it at the German Physical society in December 1900. At the time he thought about it as a "purely formal assumption" that he didn't give much thought to. And Planck was not alone.

Although his radiation law was quickly accepted by the scientific community, nobody at the time seemed to grasp its profound significance, and very few physicists expressed any interest in the justification of Planck's hypothesis. Among them was a young Albert Einstein, which was the first to realize that Planck's introduction of the concept of a quantum of energy will have profound impact on the whole physics and not just on the black-body radiation problem.

Einstein started to think that maybe what we think is continuous, like energy, actually it is not. It might be made up of a very large number of very small, but finite, quanta. Like energy, even radiation, according to Einstein, might be quantized. And this could explain another observation that was puzzling physicists at the time, the photoelectric effect. But more on this, in the next chapter.

2.6 The black-body: 3 takeaways

Everything emits light, from the Sun, to your body, to the book you are reading right now. Some of this light is detectable by the human eyes and it's called visible light. Objects at higher temperature emit more blue light, at low temperature more red light.

Physicist can measure how much light is emitted at each color (or frequency) by creating a black-body in laboratory. However, classical physics fails to explain how much light is emitted in the high frequency (ultraviolet catastrophe).

The only way to reconcile the experimental results with theory, is to assume that energy in the black-body is quantized and not continuous. The minimum quantum of energy corresponds to the Planck's constant h multiplied by the frequency of the oscillator that emits light.

2.7 To know more

- It's very instructive and entertaining to read Planck's book "The theory of heat radiation". The book is based on the lectures that Planck himself gave at the University of Berlin in 1906-07. It contains some math but the introduction and physics consideration are quite enlightening.
  - https://gutenberg.org/ebooks/40030
- For the 100 years anniversary of Planck's work that gave birth to quantum mechanics, Helge Kragh wrote a very interesting article for **Physics World**, detailing the steps to Plank's discovery.
  - https://tinyurl.com/ypn5p5ed
- If you are interested in Planck's career and accomplishments, the Max Planck Institute keeps a page with detailed Overview of Planck life and accomplishments on their website.
  - https://www.mpg.de/19252688/max-planck

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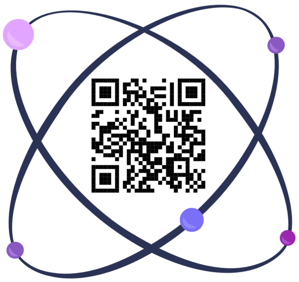

3 The Photoelectric Effect

The existence of electromagnetic waves was one of the most important prediction of Maxwell's theory of electromagnetism. The first physicist to prove his theory by creating waves in a laboratory was Hertz. But in doing so he came up with an unexpected result, at the same time challenging Maxwell's theory and shining some light on the quantum world beyond it. He discovered the photoelectric effect.

3.1 Hertz's discovery

Hertz's series of experiments to recreate electromagnetic waves in a laboratory, conducted between 1886 and 1889, were aimed at both generating and detecting waves, with a frequency lower than light, so-called "radio waves". His setup to generate waves involved creating a high voltage to cause a spark discharge between two pieces of brass. Once the spark forms, the charge rapidly oscillates back and forth. This in turn causes the emission of electromagnetic radiation, as predicted by Maxwell's theory. This apparatus was the so-called wave transmitter.

Hertz's second step involved detecting the generated waves. His idea was to build a "receiver" of the radiation, made of a copper wire bent into a circle, with a small gap between the ends, and placed in proximity of the wave transmitter described above. Once the waves hit the receiver, if Maxwell's theory is correct, an oscillating current must be created in the wire. The presence of oscillating charge in the receiver is then detected by observing a small spark between the two ends of the wire. Hertz used an adjustable thumbscrew with its end close to the opposite electrode, to precisely measure the spark length and thus measure the intensity of the electromagnetic wave.

The experiment was very successful. Not only he was able to create and detect the radiation, but he was also able to measure its properties, like polarization, reflection and refraction, confirming its nature as a wave. But the most important discovery came when he tried to improve the spark's visibility in order to make more precise measurements.

He started to enclose the receiver spark in a dark case to make it easier to observe the light. What happened then was that the maximum spark-length became much smaller than before. He then traced down this effect to a specific

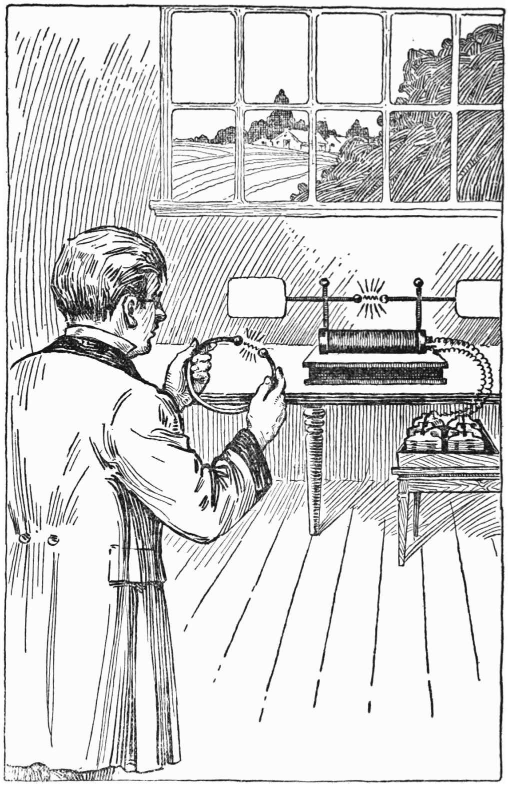

Figure 3.1: *Artistic representation of Hertz discovery of radio waves. The wave generation apparatus creates a spark between two brass balls. Each spark excites oscillating radio frequency currents, which were radiated as electromagnetic waves (radio waves). Hertz is standing, holding the receiver, made of a simple loop of wire with a narrow gap between the ends. Source: The Complete Radio Book, Yates and Pacent.*

portion of the case, the one that screened the receiver spark from the transmitter spark. Digging deeper, Hertz came to the conclusion that the receiver spark was more intense if it was exposed to ultraviolet light from the transmitter spark.

This investigation took a long time, several months. In the end, Hertz decided to stop his experiments and just communicate the results to the scientific community, without attempting any explanations or theory about the observed phenomena but leaving the problem open for investigation by other physicists in the field.

3.2 Thomson electrons

It took two decades to bring clarity to the observation made by Hertz. It all started in 1897. Thomson had been working hard on the problem of the so-called “cathode rays”. Those were rays produced by two metal ends placed in a vacuum tube, with one of them connected to positive, and the other to negative voltage. Before Thomson clarified the nature of those rays, over the years various experiments had measured their properties.

- Cathode rays were emitted from the negative charged side.
- They traveled in straight lines.
- They carried a negative charge.

- They could be deflected by an electric and a magnetic field.
- They could cause fluorescence (when they hit phosphors they emit light).

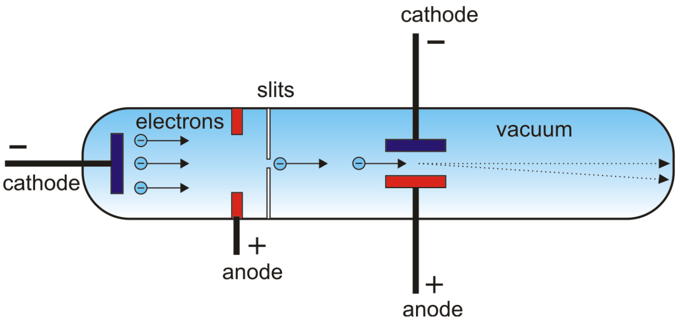

Figure 3.2: *A vacuum tube representation. Electrons are emitted from the cathode (on the left) and travel towards the anode with high speed. They are deflected by an electric field placed beyond the anode and the deflection direction and magnitude depend on the direction and intensity of the electric field. Source: periodni.com*

Despite the various properties observed, it was still not clear if cathode rays were actually particles or something of a different nature, like a wave. Thomson's series of experiments proved conclusively that those were actually particles, and he was also able to measure their electric charge and mass. Most importantly, he found that those particles could be found in atoms of all elements, and their properties didn't depend on the particular elements they were emitted from. He called these "corpuscles". Today we call them electrons.

Thomson studies of the electron continued in the next years and in 1899, when studying the experiments conducted by Hertz, he was able to establish that the ultraviolet light caused electrons to be emitted from the receiver. What happened was that the atoms on the receiver contained electrons, which were shaken by the oscillating electric field of the incoming radiation. As the radiation kept hitting the matel, some of those electrons would receive enough energy to be shaken loose, and would be ejected from the metal. These electrons were then responsible for the increase in the intensity of the observed receiver spark. Hertz had discovered the photoelectric effect.

3.3 Lenard investigations

In 1902, Lenard decided to conduct a series of experiments to investigate the properties of electrons emitted by a metal plate when ultraviolet light is directed on it. As those electrons are the result of absorption of light they were then called photoelectrons. The main purpose of Lenard's experiment was to measure how the energy of the photoelectrons changed by changing the intensity of the light shone on the metal plate.

His experimental setup was quite simple. Two metal plates were placed in a vacuum tube, and light was shone on one of the plates. The ejected electrons from one plate hit the second one, the collector. The collector plate was connected to an instrument (an ammeter) to measure the current produced by the electrons liberated by the light. This allowed Lenard to measure the quantity of electrons emitted by the plate, but not the energy. In fact, the electrons emitted by the metal plate had a certain energy, or to be more precise a certain kinetic energy, in other words a certain speed. The higher the speed the fastest they would reach the collector plate.

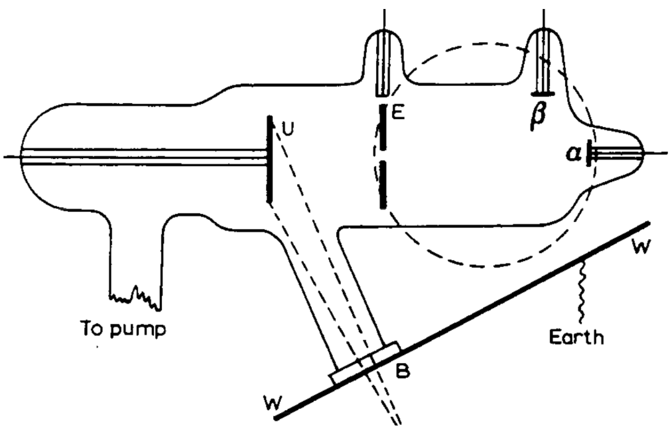

Figure 3.3: *Setup of Lenard's experiment. Ultraviolet light is shone on the plate U. Electrons are ejected, and a narrow beam is separated by the plate E. The electron beam hits the small plate α and a current is detected by an ammeter. Source: Lenard Nobel lecture.*

To measure the energy of the electrons, Lenard charged the collector plate negatively, to repel the electrons coming towards it. This way, only electrons ejected with enough speed to balance the force that repels them would hit the collector plate and contribute to the current. His idea was to increase the charge on the collector plate (also called the voltage) until no electrons started to get through (measured as zero current by the ammeter).

At this point, he concluded that the voltage value corresponded to the maximum energy of the electrons emitted by the metal, for a specific intensity of the light. He expected the experiment to go something like this:

1. Set an initial intensity of the light.
2. Observe a current with the ammeter.
3. Increase the collector plate voltage until the current goes down to 0.
4. Note down the voltage value.
5. Increase the intensity of the light.
6. Repeat from point 2.

In this way, he could find the dependency of the photoelectron energy on the intensity of the light. He expected that, as the intensity of the light increases, the electrons would absorb more energy and this would be transformed into kinetic energy by the electrons in the metal. But things didn't turn out this way.

First, Lenard discovered that there was a minimum voltage that stopped any electrons getting through, but this voltage didn't depend at all on the intensity of the incident light. As the intensity of the light increased, the number of electrons emitted also increased, but the energy of the emitted electrons staid unaffected. The more powerful waves ejected more electrons, but the individual energy of the ejected electrons was always the same as for the weaker field.

This was the first observed result in complete contradiction with Maxwell's theory of electromagnetism and with the way everybody thought energy interacted with matter. The commonly accepted view at the time in fact was that energy coming from waves could be transferred completely to matter. For example, waves cause boats to move up and down. In this sense, the energy carried by the wave is transferred to the boat and the stronger the wave, the greater the movement of the boat. But this clearly was not the case for light and electrons. Increasing the intensity of the light, didn't increase the speed of the electrons emitted by the metal.

The second surprise came when Lenard decided to explore the behavior of the photoelectrons at different colors of the light. He was able to separate the colors of his lamp and shine on the metal plate with light of different colors. He found that the maximum energy of the ejected electrons did depend on the color of the light. High frequency light caused electrons to be ejected with more energy. Moreover, Lenard found out that even light with very low intensity, but high frequency, caused some electrons to be ejected from the surface almost immediately.

This was still a fairly qualitative conclusion. Lenard wasn't really able to accurately reproduce his results, as he found it difficult to obtain strong vacuum conditions and avoid oxidation of the metal plates. But the results were conclusive, in the sense that there was a measurable dependency of the photoelectrons' energy on the frequency of the light shone on the plate. This observation cannot be fully explained using Maxwell's theory, as the frequency of the light should not play a significant role in increasing the energy of the electrons, and for very dim light it should take quite some time for electrons to accumulate the energy necessary to escape the barrier of the metal surface.

Lenard published his findings but couldn't find a reasonable explanation of the phenomena he had observed. And it really took a genius intuition to understand what really was going on.

3.4 Einstein's intuition

Despite the efforts of the entire scientific community to reconcile these experimental facts with the classical theory of radiation and matter, it was a patent clerk that had the right intuition that was able to explain all the observations related to the photoelectric effect, and more. In 1905 Albert Einstein introduced the revolutionary idea that light was not a continuous wave of energy, but it was made by a finite number of small particles that can interact with matter like the electrons in a metal, according to the same laws that govern the interaction of matter, but with an added twist.

In the introduction to his paper about the photoelectric effect, Einstein highlights the alarming state of classical physics at the time. He points out that the Maxwell's theory of electromagnetic processes differed very much from the theory of matter and gases, that are based on Newtonian mechanics.

In fact, the state of a material body is completely determined by the positions and velocities of the particles (atoms, molecules, electrons…) that constitute them. These particles can be many, but all bodies are composed of a finite number of them. In other words, matter is a discrete system. By contrast, electromagnetism describes the state of a region of space using continuous functions. For this reason, an electromagnetic state cannot be determined by any finite number of variables. This leads to the contradicting conclusion that the energy of electromagnetic phenomena (such as light) is described by a continuous function, while the energy of a material body is represented by a sum of the properties of its constituting atoms. The energy of a material body cannot be divided into arbitrarily many, small components. However, according to Maxwell's theory (or any wave theory), the energy of a light wave emitted from a source is distributed continuously over an ever larger volume. This dualism between particle and field troubled him, because it was not obvious at all how to bring those fundamental theories together in a single theory of Nature.

Einstein was very familiar with the experiments that seemed to contradict Maxwell's theory of radiation, especially in relation to the interaction of light with matter. In particular, he was very familiar with the work by Planck, which influenced his work on the photoelectric effect. Expanding on Planck's treatment of the black-body radiation, Einstein introduced the hypothesis that not only the energy of the oscillators that emit light is quantized, but that light itself is quantized, and matter can only absorb or emit light in multiples of a quantum of energy. And this quantum is proportional to the frequency of the light and the same constant h that Planck had introduced in his treatment of the black-body radiation. The photoelectric effect can then be explained as a quantum of energy that penetrates the surface layer of the solid, and is then transformed in kinetic energy of electrons. The electron then travels through the solid and, once he reaches the surface, he needs to overcome a barrier (the strength of which depends on the solid) and loses some energy.

In his paper Einstein refers to Lenard's experiments and calculates the energy of the electrons and the voltage needed to stop them from traveling from the receiver to the emitter plate at a specific frequency. His calculations turn out to be in broad agreement with Lenard's results. This argument explains the independence of the electron energy from the intensity of the incident light, since increasing the intensity would have the only effect of increasing the number of electrons that absorb a light quantum, and not their energy or speed. At the same time, Einstein's law predicts that increasing the frequency of the light would increase their energy and thus the voltage needed to stop them, proposing a way to verify his law by measuring the electrons speed as a function of the frequency of the light shone on a plate.

Einstein's theory was considered almost as a joke at the time. It was so at odds with everything that physicists knew about Nature, and even Planck had not dared to go as far as quantizing the energy itself. In fact, he considered his quantization of the oscillators' energy a mere mathematical trick, that had no physical meaning. Moreover, Einstein's theory was a direct attack to Maxwell's theory of electromagnetism. And a number of physicists at the time started to work hard, with the main goal to disprove Einstein's explanation of the photoelectric effect.

3.5 Millikan's confirmation

It took a decade of difficult experimentation before Einstein's law was fully confirmed. And it was the American physicist Millikan, who did not accept

Einstein's theory and worked for a decade to disprove it, who was forced to give up and confirm it. Millikan himself writes about the "astonishing situation that after ten years of work... this equation of Einstein's was found to predict accurately all the facts which had been observed".

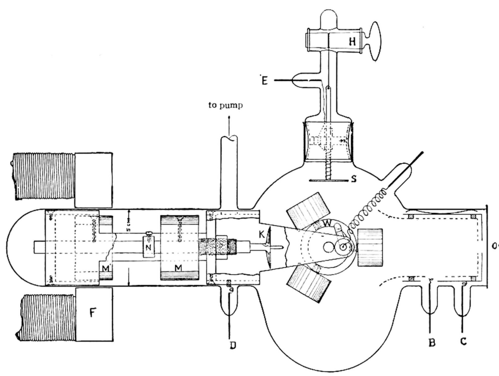

Figure 3.4: *A section of Millikan's experiment. Cast cylinders of alkali metals were placed on the wheel W. Fresh clean surfaces were obtained by cutting shavings from each metal in an excellent vacuum with the aid of the knife K. The freshly cut surface was turned so that it was opposite to point O and then a beam of light was directed towards the surface. Source: A Direct Photoelectric Determination of Plank's "h"- Millikan.*

Millikan's experiments at the Ryerson Laboratory aimed at testing the correctness of Einstein's law. The whole setup was quite similar to Lenard's experiments, with two important differences, that addressed the main limiting factors that prevented Lenard from obtaining a precise relationship between the frequency of the light and the energy of the photoelectrons.

- The range of frequencies to which metals responded was not broad enough to explore the entire possible energy of the photoelectrons, as it was limited to ultraviolet light.
- The metals in the vacuum chamber where accurately polished but, once placed in vacuum, oxidized very fast, making it more difficult for photoelectrons to escape the metals and thus altering the measurements significantly.

Millikan solved the first issue by using alkali metals as a source of electrons. In order to test the linear relation between the energy of the escaping electron and the light it was necessary to use as wide a range of frequencies as possible. While electrons are thrown from the ordinary metals only by ultraviolet light, alkali metals like sodium, potassium, and lithium respond to light frequencies from the red up to the ultraviolet.

The oxidation issue was instead solved by devising a mechanism to constantly cut shavings off the surface of the metals inside the vacuum chamber itself. This made it possible to obtain a very polished metal surface just before the light hit the metal, and minimized the issues related to the metal oxidation. With this method he was able not only to confirm Einstein's law, but also to precisely measure the value of Planck's constant, finding it in complete accordance to his first estimation from the black-body radiation law.

Einstein theory of the photoelectric effect now starts to get traction. Niels Bohr will later use Einstein's ideas on light emission and absorption in quanta to shine some light on another strange effect observed at the time, the spectral lines of excited atoms. We'll discuss this in detail in the next chapter.

3.6 The photoelectric effect: 4 takeaways

Maxwell's theory of electromagnetic radiation is very successful explaining how lights is created and propagate. But a few experiments about the interaction of light with matter showed some limitation of this theory.

The photoelectric effect (discovered by Hertz and studied by Lenard) shows that a beam of light shone on a metal plate causes the metal to emit electrons, which we call photoelectrons. Those electrons are ejected from the metal at an energy that is not proportional to the intensity of the light beam, and it seems to depend entirely on the frequency of the light. This result contradicts the prediction of Maxwell and everything we know about the classical theory of waves.

Einstein introduces a bold hypothesis to explain this effect. After considering Planck's quantization of the oscillators' energy in the black-body model, he extends the quantization concept to light itself. Einstein predicts that light is made of energy quanta, proportional to its frequency and the Planck's constant. These quanta can be absorbed by photoelectrons and the higher the frequency, the higher the energy of the photoelectron.

Millikan efforts to disprove Einstein's hypothesis end up in a fiasco. Millikan experiments clearly show the correctness of Einstein's law, decisively confirm the light quanta hypothesis and provide an accurate measure of Planck's constant.

3.7 To know more

- You can read the original account of JJ Thomson discovery of the electron in his article from 1897. This article is a very enjoyable read and provides a very clear and logical explanation of the state of physics research at the time and the struggles to explain the various phenomena observed, using only the classical theory of electromagnetism.

  - https://tinyurl.com/5c5sr8b3 (page 293)

- Lenard gave a beautiful account of the history of cathode rays in his Nobel Prize lecture.

  - https://tinyurl.com/4yn8x75s

- Millikan devoted almost all his entire scientific career to the study of the nature and properties of the electron. His book on the property of electrons is a rare gem.

  - https://gutenberg.org/ebooks/73297

- Einstein's original paper on the photoelectric effect was written in German, but luckily you can find an English translation here. It's a bit technical, but it explains well the reasoning behind Einstein's solution and his thought process about the quantization of energy.

  - https://tinyurl.com/4889kf2d

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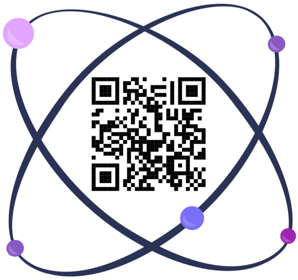

4 Atomic models

In 1814 Joseph Fraunhofer, a German lens-maker and optical scientist, is the world's finest craftsman in optical glass. While he was testing a new prism, looking as usual at the sunlight that goes through and how it creates a spectrum (the rainbow produced when white light is passed through a prism) he notices that this rainbow is divided by dozens of fine black lines. Fraunhofer was the first to see the lines, because his prism was the first one large enough and of the quality needed to notice those small details. The lines in the solar spectrum are a complete mystery to physicists, and they will stay so until Niels Bohr, a young scientist inspired by the work of Einstein and Planck, proposes a model of the atom based on the quantization of energy.

4.1 Atomic Spectra

During the 19th century a large series of experiments had established that matter was composed by a collection of very small parts, which were referred to as atoms. These atoms were thought to be the constituent parts of matter, and it was already clear to chemists and physicists alike that even atoms had some kind of internal structure. In other words, they were not elementary particles, but they were made up of even smaller parts. A lot of models were proposed, all based on classical physics, but they all had a few problems. The most important of them, they failed to explain atomic line spectra.

Over the first half of the 19th century, various chemists and physicists had been studying the light emitted by various substances, and noticed that many gases elements heated to incandescence emit light not in a continuum spectrum, or in other words they do not emit light of all colors, like a black-body for example. Instead, they emit lines of different colors, or in other words different frequencies. Every element was found to produce a unique set of lines. This highlighted that the atoms that made up each element were associated to a unique set of frequencies. Every atomic model had to account for this property somehow.

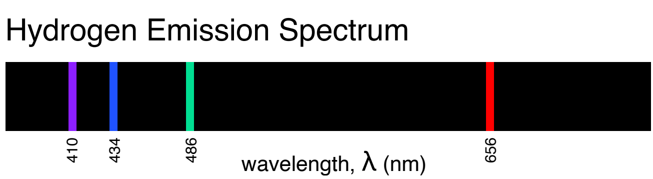

Figure 4.1: *The line spectra of the hydrogen atom. There are discrete wavelengths/frequencies of emission above a zero background.*

Numerous atomic models were proposed during the years to try to explain the experimental observations, but they all ended up contradicting the experimental evidence in one way or the other. To be fair there was really little evidence of an internal structure of the atom that could help to build a model consistent with the experimental observations. So all these models ended up being mostly speculations based on the general idea that there might exist an elementary particle that combines in different numbers and configurations to form all atoms.

These speculations ended with the discovery of the electron by Thomson in 1897.

4.2 Thomson's pudding model

Thomson's original article about the discovery of electron contains a few paragraphs where he outlined a generic idea of how those electrons could combine to form atoms. He used the theory of Prout as a starting point.

Since the beginning of the 19th century, chemists had started to investigate the properties of various elements, and were able to identify many substances and measure their atomic weights, mainly by studying chemical reactions and measuring the ratios in which different elements combined to form other substances. Studying the available data on atomic weights at the time, Prout had observed that the atomic weights of elements were all multiples of the weight of the hydrogen. This observation led him to the hypothesis that all atoms were simply made up of hydrogen atoms bound together. In other words, in his model, hydrogen was the fundamental particle from which all elements were formed. According to this theory, every other atom should have had an atomic weight that was an integer multiple of the hydrogen atomic weight. Unfortunately, as more precise atomic weight measurements started to be made, Prout hypothesis turned out to be wrong. For example, Chlorine's atomic weight was measured to be 35.5 times the hydrogen's one. This invalidated Prout's hypothesis, as 35.5 is not a whole multiple. Despite being wrong, Prout's intuition had a big influence on early atomic theories, including Thomson's.

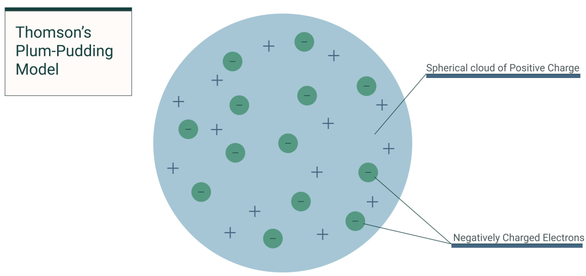

Figure 4.2: *A visualization of Thomson's atomic model. Thomson imagined the atom as a group of electrons (negatively charged) immersed in a uniform positive charge. Source: Breaking Atom*

Thomson's model started from the observation that electrons (which Thomson calls "corpuscle") have a mass lower than any known atom. This makes them good candidates to be one of atom's constituents. The second property of the electron is that it carries a definite charge of negative electricity. Since all atoms have a neutral charge, there must exist some other type of matter in the atom, with an equal charge of positive electricity. Thomson's hypothesis is that this positive charge is spread uniformly in a sphere that surrounds the electrons and has a total electrical charge equal to the sum of the electron's charges.

The electrons in the Thomson's model are not at rest, but orbit around the center of the sphere with a certain velocity, and their positions depend on the number of electrons inside the atom. Thomson attempted the calculation for an increasing number of electrons, and found out that, if the number of electrons is small, they would arrange themselves on the surface of a sphere. But, as soon as the number of electrons is higher than eight, this arrangement becomes unstable. Above this number the electrons break up into two groups. One small group arranged on a surface inside the sphere, a larger one on a sphere of higher radius. And as the number increases further, the groups become even more and the electrons arrange themselves on three or more concentric shells. With any considerable amount of electrons the problem becomes too complex to be solved with a calculation, and this is still true as of today. The Thomson problem has exact solutions only for a few values of the number of electrons (below 24).

Thomson's model was quite popular in the early 1900s as it provided an explanation of the neutral charge of the atom, and fit perfectly with the existence of the electrons. Moreover, the arrangement of the electrons in the atom was thought to somehow explain the lines observed in atomic spectra. But it ultimately failed the test of time, when Rutherford discovered that the atom is made of a hard, dense, positively charged core of particles.

4.3 Rutherford's solar system model

The first evidence that there was something wrong with the Thomson atomic model came from a series of experiments involving α particles, conducted by Rutherford and his team. The α particles were first identified by Rutherford as a positively charged radiation emitted spontaneously from certain radioactive elements. These particles turned out to be very penetrating and quite useful in probing the inner structure of the atom.

In 1909, Geiger and Marsden describe an experiment in which they used α particles, produced by a radioactive source, and directed towards different metal foils. The goal of the experiment was to find out how those particles were deviated by the electrons and the positive charged sphere present in the atom, according to Thomson's model. What they found was surprising.

A large part of the particles behaved as expected, passing through the metal foils with minimal changes of direction. But a significant part of them were turned back at very large angles. In other words, they were reflected by the atoms in the metal. This amount of reflection was small, but definitely too large to be explained by an atom where the positive charge is distributed in a sphere much larger than the α particles themselves.

Rutherford was extremely impressed by the results, and came up with the theory that this large angle deflection was due to a single atomic encounter with an intense positive electric field. In other words, sometimes the α particle got very close to a strong positive electric field, that caused it to bounce back. Rutherford's calculations showed that this sphere of positive electricity must also be much smaller than the theoretical dimension of the Thomson's model positively charged sphere. Ultimately, according to Rutherford, the atom should be made up of a strong positive central charge concentrated in a small sphere, and surrounded by a negative electric field distributed in a sphere around 10000 times larger than the central one.

This atomic model, proposed in 1911, depicted the atom as similar to the solar system, with a tiny, dense, positively charged core called the nucleus,

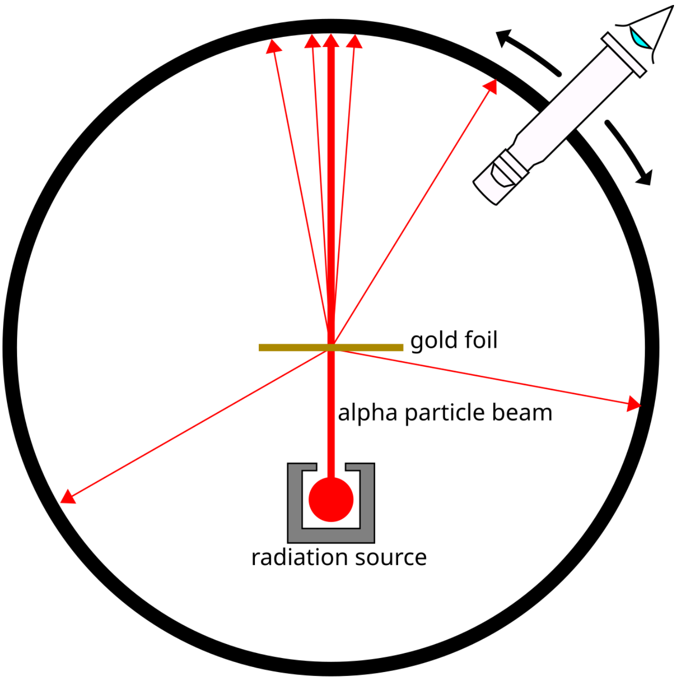

Figure 4.3: *In Geiger-Marsden experiment the α particles were emitted from a radioactive source. These particles were aimed at a very thin foil of metal, such as gold. A scintillation detector was placed around the foil to detect alpha particles scattered at different angles. Source: Kurzon, Wikipedia.*

and electrons orbiting this nucleus at some distance, similar to planets orbiting the Sun. With the atom itself made mostly of empty space between the nucleus and the orbiting electrons.

It was very clear that such a model was not theoretically possible according to classical physics. Even considering a simple system made of a positively charged nucleus of very small dimensions and just one electron describing closed orbits around it. In this simple system, the electron will start to emit radiation and lose energy, according to Maxwell's equations of electromagnetic theory. As it loses energy, the electron will no longer describe stable orbits, and it will spiral into the nucleus in orbits of smaller and smaller dimensions, with greater and greater speed. Moreover, the energy radiated out during this process, would be much higher than the energies observed in the known molecular processes.

This hypothetical system was specifically in contrast with two fundamental facts. Firstly, atoms seemed to have completely fixed dimensions, which is incompatible with the hypothesis of decreasing orbits. Secondly, in ordinary molecular processes it was always observed that the radiation process does not change the structure of the atom. And after a certain amount of energy is radiated out, the atomic system settles down in state of equilibrium, in which the distances of the particles are of the same order of magnitude as before the radiation process happened.

Rutherford was well aware of those issues and tried to fix his model by borrowing the idea of a “Saturnian” system from Nagaoka. In this model all electrons of equal mass are arranged in a circle around a dense nucleus and repel each other. Such a system could be stable in theory, if the electrons moved at the right speed. However, even Nagaoka’s model ultimately failed to explain experimental observations related to atomic spectra. It was Niels Bohr who eventually came up with the solution, by relying once again on Planck’s quantum idea.

4.4 Bohr’s quantum hypothesis

Invited to Manchester by Rutherford, Bohr worked closely with him on an improvement of the solar system atomic model. His main idea was to incorporate Planck’s and Einstein’s discoveries about the quantization of energy and to introduce the idea that electrons formed fixed, quantized orbits around the nucleus, instead of continuously spiraling towards its center.

According to this model, electrons orbit the positively charged nucleus in certain allowed circular paths called orbits or shells, each with a fixed energy level. The energy levels are represented by quantum numbers ($n = 1, 2, 3, \ldots$), and the lowest energy level closest to the nucleus is called the ground state. The ground state is the state of

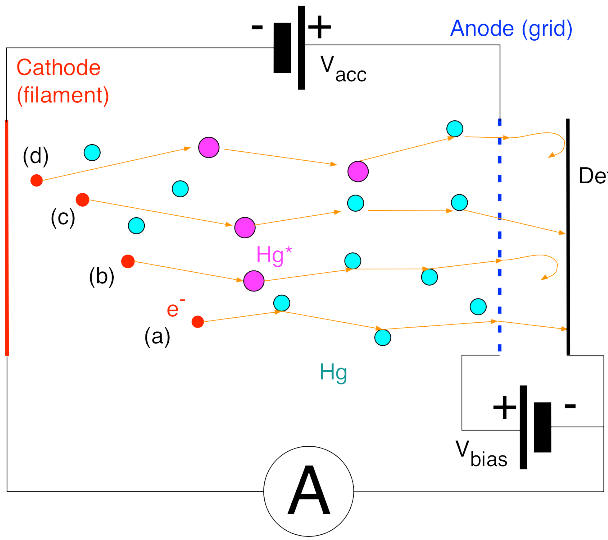

Figure 4.4: *An illustration of the Franck-Hertz experiment. The electrons emitted by the heated cathode are accelerated towards the grid by a variable voltage. Electrons crossing the grid are then slowed down by a decelerating potential on their way towards the anode. If the electron in the path inside the vacuum tube hits a mercury atom and loses energy, it will slow down and will not reach the detector. This will be seen as a drop in the current, corresponding to the energy at which the electrons of the mercury atoms can jump orbits. Source: Werner Boeglin, Florida International University.*

minimum energy of the system. In this state the electron cannot emit radiation because it would be lower than the minimum quantum of energy. If an electron can only be in specific energy levels, then it can only radiate energy by moving between these levels. The energy differences between these levels will correspond to specific frequencies, which are the ones observed in the atomic spectra. You can see that the in this way Bohr has solved the radiation problem. In fact, according to classical physics, an electron spiraling towards the nucleus will form smaller and smaller circles around it, constantly accelerating and emitting radiation at all frequencies. But this is in complete disagreement with the observation that atoms emit radiation only at specific frequencies, that look like lines in the light spectrum. Bohr's calculations bases on his atomic model show that it's possible to reproduce the position of these lines, knowing only the measured mass and charge of the electron. These calculations showed a very good agreement with the measured values of the line spectra of the hydrogen atom.

Bohr formulated his model in 1913. Just one year later, Franck and Hertz conducted an experiment, that conclusively demonstrated the existence of discrete energy states in atoms, as predicted by Bohr. As it happens very often, they almost stumbled upon this discovery. Their goal was to better understand electron conductivity through gases, and conceived the experiment years before the atomic theory of Bohr. By the time they built their experiment, collected their data and reported their results, the Bohr model of the atom was about one year old.

The experiment setup involved creating a stream of electrons by heating a metal filament inside a vacuum tube filled with very rarified mercury vapors. These electrons were then accelerated towards a positively charged plate, travelling through an electric field inside the vacuum tube. Once they reached the charged plate they had a certain energy, that was balanced by a second negatively charged plate that slowed down the electrons. The number of the electrons reaching the second plate would depend on the energy they had when they reached the first plate. If nothing was between the filament and the first plate, the electrons would have an energy corresponding to the strength of the electric field created by the first plate. But of course there were some mercury atoms that the electrons had to travel through, and the electrons occasionally interacted with them. Franck and Hertz observed that, due to these interactions, the electrons lost energy. When they were accelerated mildly, by a weak electric field, they didn't lose any energy. As the strength of the field increased, they also increased their speed, up to a certain value of 4.9 Volts. At this value, Franck and Hertz noticed the electrons speeds dropping almost to zero.

This was explained by the fact that 4.9 Volts is exactly the smallest amount of energy that the mercury atom can accept. In Bohr's model, it's the difference between the energy of one of the atoms orbit and the next available one. At first, when the electrons have an energy lower than this threshold, nothing can happen. Once the threshold is reached, an electron that comes in the vicinity of the atom can transfer its energy to it, kicking one of the atomic electrons to an outer orbit. The electron that hit the atom loses all its energy, and cannot reach the second plate anymore. This explains the decrease in electrons reaching the second plate, observed by Franck and Hertz as a drop in the current going through it.

The experiment showed that these current losses happened not only at 4.9 Volts, but also at multiples of this value. This further confirmed the theory that the electron absorbed by the atom could be absorbed again, but only if he had sufficient energy left, corresponding to the characteristic minimum absorption energy. For example, an electron accelerated to an energy of 10 Volts can hit an atom once, and transfer it 4.9 Volts. At this point he will have an energy of 5.1 Volts, it will start to travel again through the tube, and it will have a chance of hitting a mercury atom again, losing another 4.9 Volts. At this point the electron will have only 0.2 Volts of energy left. It will travel to the first plate, but it will not have enough energy to reach the second plate. This will contribute to another drop in the current observed in the second plate.

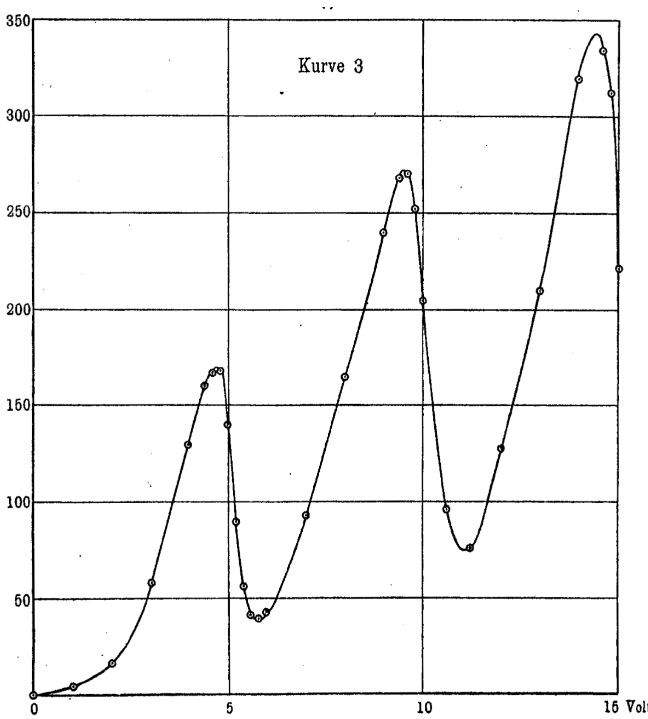

Figure 4.5: *The graph of the current dropping at different energies of the electron. The first drop occurs when the electron has an energy equivalent to 4.9 Volts. The second and third drop correspond to the case when an electron hits two or three mercury atoms. Source: Franck's Nobel lecture.*

Bohr was able to improve Rutherford's atomic model and explain many experimental observations, by simply applying the quantization of energy to the orbits of electrons around the nucleus of the atom. A few other physicists, like Sommerfeld, followed in his path, improving his model and introducing elliptic orbits, rotation around tilted axis and so on. However, the real breakthrough arrived when a new generation of physicists completely broke away from this approach, still anchored in classical physics, and instead introduced a series of new ideas, which mark the start of modern quantum physics. We'll devote the next chapter to explain their efforts.

4.5 Atomic models: 4 takeaways

Several atomic models were proposed during the course of the 19th century, but most of them were based on speculations and lacked experimental evidence. When Thomson discovered the electron, it became clear that every model needed to take that into account. Thomson's model proposed that the electron was immersed inside a positively charged sphere that constituted the largest part of the mass of the atom. But the limitations of this model were highlighted by the experiment of Geiger and Marsden, who discovered that the atom is made of a small, hard central core.

Rutherford proposed a model similar to the solar system, with the atom made of a central positive charge, surrounded by electrons orbiting around it. This model was in agreement with the experiments of Geiger and Marsden, but it had many issues. First, according to classical physics, it was difficult to explain its stability. The electrons orbiting around it should have lost energy and ultimately collapse inside the nucleus, but this was clearly not the case.

Bohr improved Rutherford's model, postulating that electrons could only occupy specific orbits. He took inspiration from the work of Planck and Einstein, that had successfully explained the black-hole and the photoelectric effect, by introducing the hypothesis of energy quantization. In the case of the atomic model of Bohr, the energy of the electron orbiting inside the atom is quantized, and each orbit corresponds to a specific energy. To get from an orbit to another, the electron had to absorb a specific amount of energy. This explained why atomic spectra are not continuous, but have lines characteristics of each atom.

Bohr's model was confirmed by the experiment by Frank and Hertz. They successfully proved that electrons travelling through mercury gas could hit the atoms and bump one of its electrons to an orbit with a higher energy. But the most important thing is that this happened only if the electron had a minimum speed, corresponding to an electric field of 4.9 Volts. This confirmed Bohr's atomic model, and the hypothesis of orbit quantization.

4.6 To know more

- This book by Thomson is a very clear account of both the experiments and the theory behind the interaction of electricity and matter. It covers a range of topics, from the classical theory of electricity to radioactivity. It contains a dedicated chapter to his atomic model, which, although the model described there turned out to be wrong, is very interesting to read.
  - https://tinyurl.com/bdedwwhn
- The wonderful Helge Kragh gives a detailed account of the theories of the atomic structure up to and including Rutherford's atomic model in this book. Before Bohr: Theories of atomic structure 1850-1913
  - https://css.au.dk/fileadmin/reposs/reposs-010.pdf
- Geiger-Marsden original article on their famous experiment is reproduced here. Exceptionally clear and very interesting to read.
  - https://www.chemteam.info/Chem-History/GM-1909.html
- Rutherford's article on The Structure of the Atom is reproduced here. He outlines his model but also points to the fact that it's difficult to justify, based on classical physics

only. It also points out that Niels Bohr model was working on an interesting attempt to reconcile the ideas of Planck with a coherent atomic model theory.

  – https://www.chemteam.info/Chem-History/Rutherford-1914.html

- Niels Bohr's series of articles on his model of the atom. He covers the structure of simple atoms and accurately predicts the spectrum of the hydrogen atom.

  – https://gutenberg.org/ebooks/72787

- Franck and Hertz received a Nobel Prize for the experiment that confirmed Bohr's theory. In this lecture Franck outlines the experiment, its conclusions and findings in great detail.

  – https://tinyurl.com/yk8286mz

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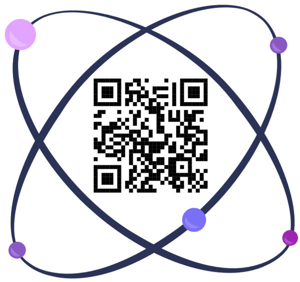

5 A New Physics

The efforts of Planck, Einstein and Bohr over the first two decades of the 20th century, had brought a great deal of insights and significantly advanced the understanding of microscopic phenomena that were being discovered by physicists all around the world. However, the quantum theory was still a phenomenological one, in the sense that it was able to describe experimental facts, but lacked a coherent structure, which is what characterized classical physics like Newton's theory of gravitation or the theory of electromagnetism described by the Maxwell's equations.

But the time was ripe for a new theory, and in 1926, almost at the same time, two brilliant physicists came up with two radically different formulations of what we call today quantum mechanics.

5.1 De Broglie's waves

As we saw in the previous chapter, in the early 1920s the entire physics community was occupied by two striking facts, confirmed by numerous experiments. The first was the apparent contradiction of the nature of light, that appeared to have properties typical of waves and of corpuscles at the same time. The second fact, was the property of electrons in atom. Instead of taking up all possible (in theory infinite) motions and orbits around the nucleus, they appeared to only occupy a finite number of them.

De Broglie was not the first to notice these difficulties, but he was probably the first to link the newly discovered properties of light and electrons and try to unify them in a single theory of matter and radiation. He was not satisfied with neither of the theories existing that he time. He was not convinced by the light-quantum theory (elaborated by Einstein) because it related the energy of a light particle (later to be called the photon) to its frequency. And in a purely corpuscular theory there is nothing that allows the definition of a frequency.

Regarding the electron issue, to determine the stable orbits of electrons in an atom it was necessary to use whole numbers. And the only phenomenon that involved whole numbers in physics at the time was related to the harmonic oscillations of systems. For example, a string can vibrate with a fundamental frequency, that depends on its length, and all the integer multiples of that frequency, which are called harmonics. This suggested that electrons too could be thought of systems with some kind of periodicity.

This brought De Broglie to the logical conclusion that for both matter and radiation it was necessary to consider the concepts of particle and wave at the same time. In other words, both matter (like the electron) and radiation (like the photon) had to exist as particle and wave at the same time, in all cases. Moreover, it had to exist a relation between the motion of a particle and the propagation of the associated wave.

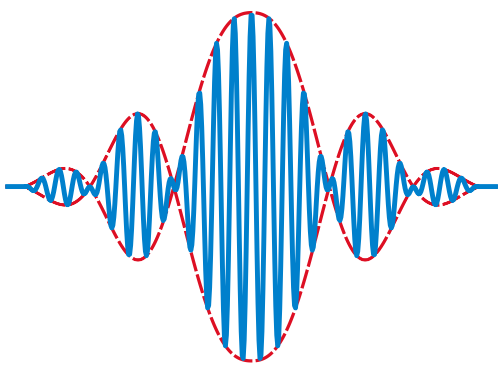

Figure 5.1: *A wave packet is a localized group or burst of waves that travel together as a unit. It is formed by superposing multiple waves of different wavelengths or frequencies, which interfere constructively over a small region of space to create a localized wave while destructively interfering elsewhere. This results in a wave that is confined in space, as in De Broglie's theory.*

De Broglie's theory unified radiation and matter, identifying each particle with a wave packet, which is a set of waves with similar frequency confined in a small space. Then, what we see in an observation is the wave packet, and the speed of the particle is defined by the speed at which this packet moves, which is equal to the frequency of the wave packet multiplied by the Planck's constant. The higher the frequency of the wave, the higher the energy and speed of the particle.

This hypothesis allowed explaining also the quantization of energy levels in atoms. By requiring that the allowed orbits corresponded to a wave propagated in such a way as to circle the nucleus, it constrained the wavelength to be a multiple of the circumference of the nucleus. And, since the energy of the electron is assumed to be proportional to the associated wave frequency, this immediately implied the quantization of the electron energy. This turned out to be exactly the same conclusion Bohr arrived at when building his atomic model, and correctly predicted the lines of the hydrogen atomic spectrum.

A natural consequence of De Broglie's theory was that the properties traditionally associated only with waves, such as diffraction for example, could also apply to matter itself. In the case of diffraction, when a wave encounters an obstacle or opening comparable in size to its wavelength, it effectively becomes a secondary sources of new spherical waves that start to propagate from the opening and interfere with each other, This creates characteristic patterns of maxima and minima in intensity.

This should then be the case also for electrons and

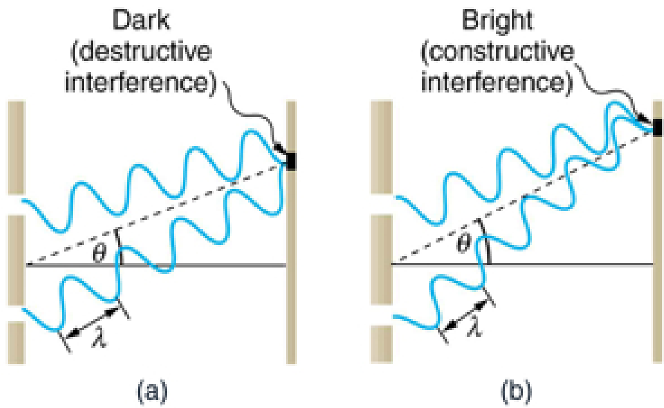

Figure 5.2: *When light passes through a double-slit or is diffracted by a crystal, it causes the waves to spread out. These diffracted waves then interfere with each other to create the characteristic bright and dark fringe pattern on a screen. Source: Lumen Learning.*

other microscopic particles, that were predicted to behave not only as discrete points but also like waves spreading out and creating patterns much like light waves do when they pass through narrow openings. This prediction was confirmed by the experiment of Davisson and Germer in 1927.

In the experiment, electrons were fired at a nickel crystal, at increasing speeds. The scattered electrons were detected at various angles and the intensity of scattered electrons was measured. At a particular accelerating voltage (around 54 V) and scattering angle (~50°), a significant peak in the intensity of scattered electrons was observed.

This result could not be explained by the hypothesis that electrons had only a particle-like behavior, because in that case the scattering should have decreased at large angles, not peak at a specific angle. The phenomenon was instead attributable to the diffraction by the crystal plane on electron waves. In fact, electrons that hit one of the atoms on the plane, and are deviated at a specific angle, will interfere with electrons from the atoms in the vicinity that are deviated in the same direction. The interference effect happens because the distance between the atoms in the crystal plane is of the same order of magnitude of the electron wavelength. The diffraction pattern observed was analogous to X-ray diffraction by crystal planes, and its intensity was in perfect agreement with the predictions by de Broglie's formula.

It's hard to understate the profound impact that

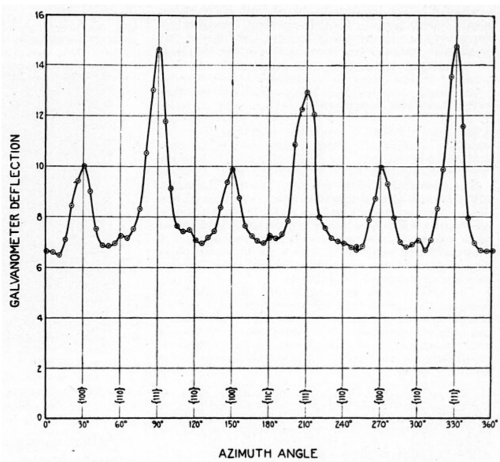

FIG. 2.—Intensity of electron scattering vs. azimuth angle—54 volts, co-latitude 50°.

Figure 5.3: *The diffraction pattern observed by Davisson and Germer. When electrons were accelerated to about 54 eV kinetic energy and directed at a nickel crystal, a distinct pattern appeared. This confirmed that electrons have wave-like properties. Source: Davisson and Germer original article.*

De Broglie's idea had on the development of Physics. The fact that to describe the properties of both matter and light, we had to introduce the concept that particles are waves, and vice versa, meant that for example an electron could no longer be thought as a single, tiny bit of matter with an electrical charge. It had to be associated with a wave, and this wave had specific properties that could be measured, like its wavelength and the typical wave-like diffraction patterns.

The Pandora box was opened, and what was up to then a collection of strange rules, used to explain phenomena that escaped the ordinary physical laws valid in the macroscopic world, started to develop into a more coherent description of the behavior of the microscopic world. From what we call now the old quantum physics, Schrödinger and Heisenberg developed a new set of laws, that describe how these wave-particles behave and propagate with time. The equivalent of Newton's laws for the microscopic world.

5.2 Schrödinger's wave mechanics

The theory of De Broglie produced a deep impression on Schrödinger. In 1926, he came up with an original formulation of the so-called wave mechanics. A theory that explained how the particle waves theorized by De Broglie propagate in space and evolve with time. His theory successfully managed to deduce both the laws of motion and the conditions of quantization of energy at the same time from basic principles.

In order to do so, Schrödinger introduced the concept of wave function. This is a concept similar to the particle wave introduced by De Broglie, but this time it's not tied to any specific physical quantity. The wave function in Schrödinger's theory is an abstract object that defines a quantum system and its state. It tells how the system evolves with time and in the simplest case of a single particle, it assigns a value to each point in space, similar to what happens with waves that propagate with time. In fact, the description of the evolution in time of the wave function is the uniqueness of Schrödinger's approach, and made his formulation a pillar of the new quantum physics.

I don't want to get into the details of the equation here, but it's important to notice how the equation doesn't specify anything about the quantization of energy, which is instead a postulate of Planck's and Einstein's theories of the interaction between radiation and matter. Even Bohr's model of the hydrogen atom introduces the quantization as an artificial postulate, on top of a very classical theory of the interaction of electrons with positive particles. Schrödinger approach is completely different. Starting from an equation which is the equivalent of Newton's laws, but applied to wave functions, he was able to obtain the energy quantization result naturally. In other words, there is nothing discrete in the wave function or the equation that describes its evolution. It's just that the solutions to the equation have a discrete nature and the energy quantization emerges from them as a result of solving it, by requiring only that the system has a definite energy that doesn't change with time.

There is nothing particularly new about it. In fact, lots of physics problems related to waves have discrete solutions. For example, a string with a definite length, will vibrate at frequencies that are all integer multiples of a fundamental frequency. This is what harmonics are. In this sense, the solutions of the Schrödinger's equation for the hydrogen atom show that the electron can vibrate around the nucleus with a fundamental frequency (or energy, according to De Broglie's relation) and integer multiples of this, corresponding to the harmonics of the fundamental energy state, the so-called ground state. And those solutions turned out to agree pretty well with the experimental observations, as well as the results from Bohr, but without introducing arbitrary quantization conditions.

The introduction of a wave function to explain the behavior of quantum systems turned out to be very useful for physicists. The Schrödinger's equation could be solved, it could be applied to a variety of physical systems, it allowed to make predictions. In short, it solved a lot of practical issues. However, it was not clear at all what the wave function

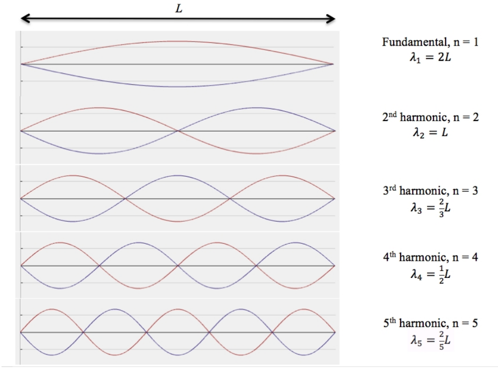

Figure 5.4: *The harmonics of a string fixed at both ends are all the wave patterns that can form in such a system. Each harmonic corresponds to a vibrating mode with a distinct frequency and wavelength. Source: AAPT.*

was supposed to represent. In classical physics, the laws of motion describe exactly what are the positions and speeds of each part of a system, at each point in time. For example, you can exactly calculate the position and speed of a cannonball at each instant, after it's being fired. Does this hold true for the wave equation as well? Does the Schrödinger equation predict the position and speed of the wave at any point in time? The short answer is no.

The equation predicts only the evolution of the quantum state, once all the forces acting on it are known. To extract a physical meaning out of it, Max Born, a German physicist who was working on a similar theory to the Schrödinger's one, introduced the probabilistic interpretation of the wave function. According to Born's interpretation, the wave function itself is not a physical wave but a function that describes the probability of finding a particle at a given point in time. This interpretation allows relating the wave function to physical and measurable quantities, like the position of a particle. But it introduces a disturbing feature of the theory. Measuring the position of a particle, will only yield a probabilistic outcome. In other words, if you prepare a system in a specific quantum state, there is no guarantee that two different measures will give the same result. In the case of the hydrogen atom, the wave function describes the spatial distribution where the electron is most likely to be found. This means that a series of measurements of the position of the electron in the atom, will give different results. The measurements results will however be distributed according to the wave function, which predicts the chance of finding the electron in a precise point in space, once a measurement is performed.

5.3 Other formulations of quantum mechanics

Schrödinger's equation is considered the easiest way to visualize and in general work with quantum systems. However, it's not the only formulation of quantum mechanics that physicists came up with. In particular, it's interesting going over two

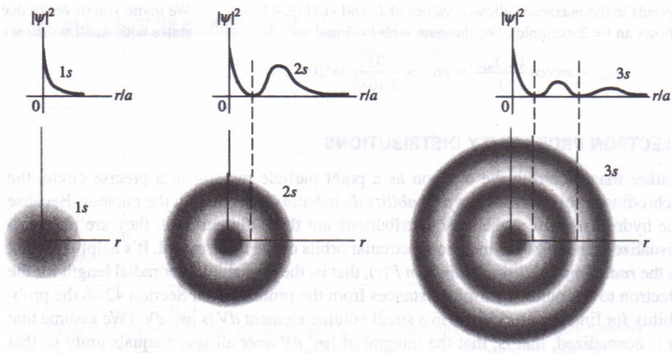

Figure 5.5: *The distribution of electrons in a hydrogen atom, for different orbitals or energy levels of the atom. The probability distributions are shown as a function of the distance of the electron from the atomic nucleus. Source: physicsmax.com.*

alternative formulations, the one from Heisenberg and the one from Feynman.

Werner Heisenberg came up with his formulation independently and actually a few months before Schrödinger. After studying the work by Bohr and his astronomical oriented theory of describing the hydrogen atom as electrons orbiting around a positive nucleus, he was convinced those ideas were not going anywhere. Instead, he developed matrix mechanics. The main idea of this formulation was to create a representation that allowed to manipulate energy and position of electron orbitals without making any hypothesis of the underlying physical process behind the measurement outcome. Since the electron orbits are not really observable anyway, Heisenberg proposed that speed and position of the electrons should instead be treated as observables. Which means they cannot be defined before making a measurement. This is the insight that allowed him to correctly predict the hydrogen spectrum.

This formulation of quantum mechanics is very useful and familiar to all physicists these days, but it required knowledge of matrix formalism, which was not that popular at the time. And that's the main reason why Heisenberg's ideas were not immediately adopted by the physics community. Schrödinger's formulation on the other hand immediately gained traction, as it made use of a more familiar formalism based on differential equations, that describe the way something changes when some amount of time passes. These days the situation is probably reversed. I studied matrix algebra from the first days as an undergraduate, so I was very familiar with it when they were introduced in my first Quantum Physics lecture.

Another formulation of quantum mechanics I personally like is the one invented by Richard Feynman. It's called the path integral formulation or sum-over-histories approach. The idea is that the probability for a particle to go from point A to point B is given by summing over all possible paths that the particle could take between those points. At each point in time, we don't really know what path the particle has chosen and, since you can't specify which way the particle goes, you sum them up.

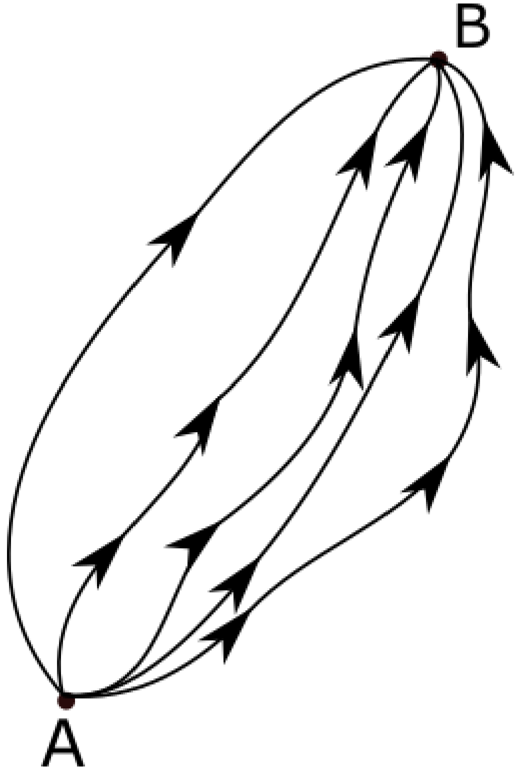

Figure 5.6: *In the path integral formulation of quantum mechanics, a particle does not have a single unique trajectory from an initial point to a final point. Instead, it takes all possible paths between those points. These are five of the infinitely many paths available for a particle to move from point A to point B. Source: Wikipedia.*

This approach turns out to be a generalization of the classical principle of least action. The principle states that among all possible paths a physical system could take to go from an initial state to a final state, the actual path taken is the one for which a quantity called the action is minimized. The action quantity is a measure of the total effort along a path, or more specifically, how the energy of a system changes in time along a particular path. The principle of least action states that nature always prefers the path with the least action, or the one where the energy changes the least during the path.

No matter the formulation, quantum mechanics finally looks like a theory that describes all those observations that had puzzled physics for almost a century. But all this came with the price of making the microscopic world (and ultimately the very foundation of Physics) governed by laws that are completely at odds with our everyday experience. We will explore those laws and their strange consequences in the next chapter.

5.4 A New Physics: 3 takeaways

Inspired by Planck and Einstein's approach to the radiation-matter interaction, De Broglie proposes a radical idea. That electrons (and particles in general) behave like waves, with their energy proportional to the frequency of the wave packet. This allows the derivation of the quantization rule for the hydrogen atom and is confirmed by the observation of electron diffraction in an experiment by Davisson and Germer.

The intuition by De Broglie serves as the starting point for one of the first formulations of quantum mechanics. In 1926 Schrödinger comes up with an equation that describes the evolution in time of a wave function, that represents the quantum state of a particle. The Schrödinger equation is the equivalent of Newton's laws of motion but, unlike in classical mechanics, the wave function doesn't describe a trajectory in classical terms. According to Born interpretation, the wave function is not a physical wave but instead describes the probability of finding a particle at a given point in time. The quantum world loses determinism.

Almost at the same time, Heisenberg comes up with an alternative formulation of quantum mechanics. His idea is to create a theory that allows to predict energy and position of electron orbitals without describing them as trajectories in the classical sense. Heisenberg proposed that speed and position of the electrons should instead be treated as observables, that cannot even be defined before making a measurement. Heisenberg and Schrödinger's theory are later showed to be equivalent.

5.5 To know more

- You can find an account of De Broglie's hypothesis in his Nobel lecture. It's an insightful account, that reveals the origin and the train of thoughts behind his discovery of the wave nature of matter.
  - https://tinyurl.com/49n2f9jt
- Davisson and Germer detailed paper on electron diffraction, which explains the experiment setup, data analysis and the curious accident that was the reason why the experiment was performed in the first place. An accident that led to the Nobel Prize.

  - https://tinyurl.com/42t3r93w

- Schrödinger's original paper on wave mechanics is a bit technical, but it contains a good amount of discussion on the derivation of his equation and how it actually has many parallels with classical theory.
  - https://tinyurl.com/42zhw9h5

- The *Umdeutung* paper by Heisenberg is the first formulation of his quantum theory. Admittedly obscure in many parts, it has been described as "pure magic" by Weinberg. Recently, a great video appeared, that makes a very good job at explaining the paper step by step. Really worth watching.
  - https://tinyurl.com/ys7uscfh

- Anything written by Richard Feynman is always entertaining as well as incredibly insightful. This Nobel lecture is no exception.
  - https://tinyurl.com/3frhtcfd

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6 Quantum Effects

The new formulation of quantum mechanics ideated by Schrödinger and Heisenberg was conceived as a fundamental theory that was supposed to describe the way objects behave, not only at the microscopic level, but at all scales. However, there are a few features of the theory that are completely at odds with our common sense and daily experience. The idea that particles are both wave and matter, the important and peculiar role of measurement in the theory, and the strange concept of quantum superposition have puzzled physicists since then.

The theory of quantum mechanics has been extremely successful at predicting a very wide range of phenomena, from radioactive decay to superconductivity. But despite being great at describing our observations, the theory itself raises a few questions on what reality is and how much can we really know about the world.

6.1 Wave-Particle duality

The Schrödinger's formulation of quantum mechanics successfully reconciles all the experimental facts that pointed at matter behaving like waves (in the Davisson Germer experiment) and wave behaving like matter (as in the photoelectric experiment). By introducing the concept of wave function we are now able to describe how a quantum system (like an electron in a hydrogen atom for example) behaves and evolves with time. But then, if everything can be described by a wave function, how do we explain the reality that things actually exist in a specific point in time and are not just distributed in space with a certain probability of being here and some being there? Even the atoms that make up this book are certainly inside this book, and not somewhere else 100 miles away. You might think that the Schrödinger equation somehow solves that. That if you let the wave function of an electron evolve it will naturally stabilize around a definite position. The equations say the exact opposite. They show that letting the electron evolve will make its wave function spread around billions of miles in a very short time. In other words, the electron will have a small probability to be anywhere in the Universe. How to solve this?

The solution lies in the concept of measurement. In quantum mechanics, the measurement process is somehow embedded in the theory. Remember Heisenberg approach? We don't really know any-

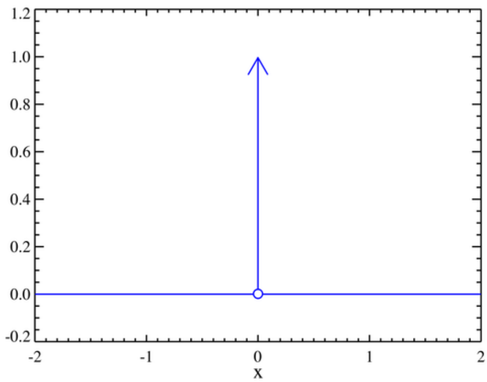

Figure 6.1: *A representation of the Dirac delta function. The value of the function is zero everywhere apart from one point. The arrow represents the area under the function. When used to describe the collapse of a wave function, this means that the probability of measuring a specific value of a quantum system observable, like the position, is 100% in one point and 0 everywhere else. Source: Omegatron, Wikipedia.*

thing about the orbits of electrons in atoms. All we know are observables, and these observables are the results of measurements. Let's make an example of such an observation. You probably know that some atoms are radioactive. What it means in practice is that they are not in a stable configuration and tend to decay to a more stable one by emitting some particles. For example, Carbon-14[1] is unstable and decays to become Nitrogen-14. When it does so, it emits an electron (plus another particle called neutrino but let's not complicate things too much). The electron emitted can be observed in a variety of ways. In the 1920s physicists used cloud chambers, that are literally chambers filled with gas. When electrons pass through the supersaturated vapor within the chamber they knock off other electrons from gas molecules along their paths. The resulting ions act as condensation centers for vapor droplets, making the electron's trajectory visible as a misty trail. A hundred years later, we use more sophisticated methods like layers of silicon. But the principle is the same. Electrons passing through matter attract other electrons inside the matter itself. And this generates a current in a specific point of the silicon, that allows to precisely determine where

[1] The nucleus of atoms is made up of particles called nucleons. There are two types of nucleons, the neutrons (that have no electric charge) and the protons (that have positive electric charge). Here the 14 means that this nucleus of Carbon has 14 nucleons, of which 6 protons and 8 neutrons. Carbon has always 6 protons, but the number of neutrons can change.

the electron was at a precise point in time.

These are all examples of a measurement, defined as the observation of a specific property of a particle. It can be the position, the speed, the energy, and so on. What happens to the wave function when we make an observation? Well, it collapses to the value corresponding to the measurement outcome. In the case of the observation of a particle's position, the wave function of the particle becomes zero everywhere apart from the actual measured position. This idealized collapsed state is modeled mathematically as a Dirac delta function, representing the particle being found exactly at the measured position. You can then think of a particle as behaving like a wave when there is no observation, and as a particle when we observe it.

6.2 The Uncertainty Principle

A necessary consequence of Quantum Physics, in any formulation, is the so-called uncertainty principle. In its original formulation, it states that it is impossible to simultaneously measure exactly both the position and the velocity (or the momentum to be more precise, but the idea is the same) of a quantum state, like a particle. The more precisely one of these quantities is known, the less precisely the other can be measured. This limitation is not due to experimental errors but arises from the inherent nature of particles and waves at subatomic scales. Mathematically, the product of the uncertainties in position and momentum is always greater than or equal to a multiple of the Planck's constant.

But what exactly happens to position and speed when we observe a quantum state, according to the uncertainty principle? When we measure the position of the electron in the previous example, we are collapsing its wave function to a point in space, so we can describe it as a Dirac-delta peaked to a single point. What happens to the speed of that particle right after the measurement? Since the uncertainty on the position is essentially zero, the uncertainty on the speed is infinite. The particle can have any speed in any direction. And the opposite happens. Suppose to have a particle that propagates with a well definite speed, without any uncertainty. This necessarily implies that the particle can be anywhere. It's a hard fact to accept, but it's a reality of nature. Physicists often interpret this in the light of Heisenberg's approach, that implies that position and speed are not real, and actually we are not interested in what is real at all. What we know is the quantum state of a system, described by the wave function and its evolution. The physical theory allows us to make predictions, that are probabilistic, about the observables of a state. Position and speed are simply ways to describe the quantum state according to quantities that we are familiar with, and that we can measure, but that happen to not be independent of each other.

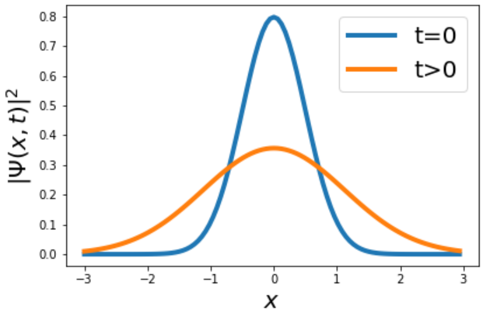

Figure 6.2: *A wave packet spreading over time. At the beginning, the packet is localized around a point. As time passes the wave function spreads more and more. For free electrons, the spread happens at incredible speed, making the probability of finding the particle far from the origin point more and more likely. Source: Delft University of Technology.*

The uncertainty principle was used by Einstein as a proof that something was wrong with the quantum mechanics formulation, or better with the Copenhagen interpretation of quantum mechanics, held by Heisenberg and Bohr. Einstein believed that the uncertainty principle stemmed from our ignorance of reality. In other words, Heisenberg's quantum theory was a good phenomenological description of reality, but could not be a fundamental theory, which was still to be found. Bohr and his disciples instead believed that uncertainty is a fundamental property of reality, and quantum states could not be described in deterministic terms. This debate went on for many years in the course of the development of the quantum theory, but it eventually faded away, together with the main creators of the quantum theory.

Nowadays, nothing of this sort is taught in Universities and these fundamental questions about the nature of reality are discarded by physicists as naive and something that philosophers would bother with. Quantum physics is a perfectly working theory, that had an incredible success over the course of decades at predicting a wide range of phenomena. Whether it describes reality or it's just a useful representation, it's not really an interesting question to the majority of modern physicists.

6.3 Quantum Superposition and Entanglement

You have probably already come across Schrödinger's cat concept. Imagine a cat placed in a sealed box containing a small amount of radioactive substance, a Geiger counter, and a small dose of sedative. The radioactive substance is so small that there is a probability of 50% that one atom will decay within an hour. If the radioactive atom decays, the Geiger counter detects it and triggers the release of the sedative, making the cat fall asleep. If the atom doesn't decay, the cat stays awake. The original version was more cruel to the cat, but I prefer to keep the cat alive and well.

According to quantum mechanics, until the box is opened and observed, the atom is in a superposition of decayed and not decayed states. This implies the cat is simultaneously both asleep and awake. This is a superposition of states. Only when the box is opened and observed does the wave function "collapse" to a definite state (awake or asleep).

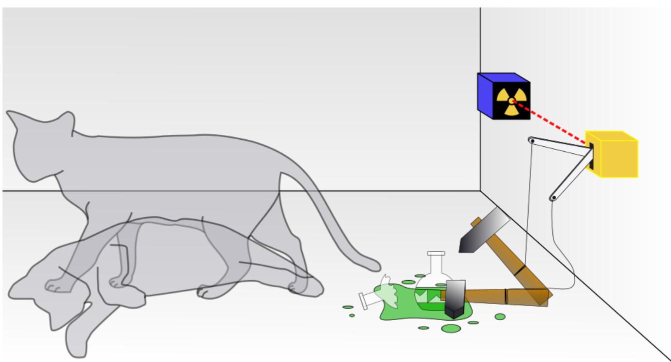

Figure 6.3: *A cat, a flask of sedative, and a radioactive source connected to a Geiger counter are placed in a sealed box. The quantum description uses a superposition of an awake cat and one that has fallen asleep. Source: Science.*

Schrödinger came up with this experiment, which I have slightly modified to make it more cat-friendly, in an attempt to point out the absurd conclusions of the Copenhagen interpretation of quantum mechanics. He argued that the measurement problem and the consequent superposition of states that happen in the microscopic world can in principle affect the macroscopic world, leading to absurd results. Ultimately his critic was that the Copenhagen interpretation considered reality as something independent of what could be established with experiments. The idea of a quantum state not really accessible and with uncertain properties (like a radioactive atom that has a 50% probability of decaying within an hour) could easily lead to absurd conclusions in the macroscopic world. However, despite being one of Einstein's favorite argument against the quantum theory, many physicists believe this thought experiment doesn't highlight any issues with it, and it can be explained in terms of two phenomena, quantum entanglement and decoherence.

We have seen how the equations of quantum physics describe the state of a system as a wave function. For example, an electron inside a hydrogen atom is in a quantum state. But what about the other things in the atom? The nucleus of the atom is made up of a particle as well, the proton. Is the proton in a separate quantum state? Not really. According to quantum physics, the quantum state that describes the system is a single one, and its wave function depends on the properties of both particles, like position and speed. In other words, the particles in the hydrogen atom are not completely independent by each other. Even if most of the time we ignore this fact, because it simplifies the calculations quite a bit, we cannot ignore it if we want to interpret what is really going on. When we have got a quantum state that is made of two or more particles, we say that these particles are entangled with each other.

An example of entanglement is the hydrogen atom, but entanglement really is everywhere. Every interaction of a particle with another can lead to entanglement. For example, let's say that an unstable particle disintegrates and emits two particles, A and B. The quantum state of the system made of the two particles moving in opposite directions is entangled. We don't know the direction and speed of particle A and particle B will move, but we know that energy is conserved, so when we measure the speed of the first one we immediately know the speed of the second one. The measurement of the first particle speed makes the wave function collapse, but the wave function is unique for both particles because they are entangled. This means that we have observed both particles at once, even if they are very far apart. This is the essence of quantum entanglement.

But what does entanglement have to do with Schrödinger's cat? Let's see the experiment with the lens of entanglement. The state of the radioactive substance is in a superposition of states, decayed and not decayed. But then the Geiger counter too is in a superposition, because its measurement depends on the radioactive substance decaying or not. And with it the mechanism to release the sedative, and ultimately the cat itself. The whole system should be entangled with the radioactive substance.

But in reality is impossible for a macroscopic system like the cat in the box to remain completely entangled with a microscopic system in a superposition state. What happens in practice is that there are many interactions between our cat and its surroundings, from the atoms of the air he breathes to the photons in the box to whatever is around him, that interact with him and eventually destroy the coherence of the system and the cat's entanglement with the radioactive atom's quantum state. This process is called decoherence, not surprisingly. The result is that the cat behaves at all items as either asleep or awake. There is no superposition of states. Despite the efforts of physicists to create a macroscopic quantum state made of many particles entangled with each other, it turns out that decoherence in macroscopic systems happens very rapidly and with near absolute certainty.

Despite being quite unintuitive and hard to reconcile with our experience of the macroscopic world, Quantum Physics is without a doubt a very successful theory, and in the decades after its formulation in 1927 it was extended and applied by many physicists to a wide variety of problems. Together with it came the first applications of the theory, and it's fair to say that the main technological developments of the past century are all a product of Quantum Physics one way or the other. More on this in the next chapter.

6.4 Quantum Effects: 3 takeaways

The Schrödinger's formulation of quantum mechanics describes particles using wave functions, but our experience tells us that matter actually exists in a specific point in time and is not just distributed in space. This is explained by the fact that measurement is a process that makes the wave function collapse in a certain point in space. You can think of a particle as behaving like a wave when there is no observation, and as a particle when we observe it.

A necessary consequence of Quantum Physics is the uncertainty principle. It states that it is impossible to simultaneously measure exactly both the position and the velocity of a particle, unlike in classical physics. This principle bothered many physicists at the time, like Einstein. While others like Bohr, were convinced that uncertainty is a fundamental property of reality. Although this debate went on for many years, nowadays, physicists tend to focus on the applications of quantum physics and accept the uncertainty principle without really questioning its deep meaning and de facto accepting the Copenhagen interpretation.

Schrödinger's cat experiment was an attempt by Schrödinger himself to highlights the absurd conclusions of the Copenhagen interpretation. In the experiment the superposition of states in a microscopic atom leads to the absurd conclusion that a cat is simultaneously awake and asleep. The introduction of the concept of decoherence helps to make sense of this clever experiment. Macroscopic systems cannot really be kept separated from the environment for too long. When they start to interact with it, they rapidly lose the initial coherence with the microscopic entangled state, and become disentangled from it. This helps explain why is impossible to observe superposition of state in macroscopic systems.

6.5 To know more

- A pretty non-technical and very clear explanation of quantum coherence can be found in this paper. It also offers a deep dive in various interpretations of quantum mechanics.
  - https://arxiv.org/abs/quant-ph/0312059
- There were a few attempts to replicate Schrödinger's cat thought experiment in

a laboratory (without using cats). This paper highlights the difficulties of observing macroscopical effects of a superposition of quantum states.

- https://tinyurl.com/55f8ejwk

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7 Quantum Physics Everyday

Since the discovery of quantum physics, more than 100 years ago, the technologies used in the early experiments were further developed and found a place outside the physics laboratories to be used in everyday life. Most of the modern technology that powers many of the devices we use every day wouldn't exist without the discovery of quantum physics and of all the quantum effects that we discussed in the previous chapters.

7.1 Microelectronics

The history of computers started a few millennia ago, with the abacus first used in Babylon more than 2000 years BCE. And over the course of history many types of mechanical devices were developed by many civilizations, for the purpose of automating and speeding up computations. For example the Antikythera mechanism, completely based on mechanical gears, and considered the first analog computer, was used in Ancient Greece to predict the movements of the stars.

A few centuries later, the discovery of electricity gave rise to new generations of programmable machines, that first used electrical and mechanical elements, like the Z3, then fully electrical elements, like the ENIAC. Those machines were built using vacuum tubes, which are electronic devices that control electric currents flow in a high vacuum. They consist of a cathode (which emits electrons), an anode (plate), and sometimes additional control elements.

Vacuum tubes like the triode were used as the basic building blocks of logical ports, that make possible the automatic computations inside a computer. But, as the needs for more computational power increased, the computers started to need more and more vacuum tubes, which led to a huge increase in the size of the computers, and an increase in the energy needed to power them. Vacuum tubes size was around 5 cm (or 2 inches) and they were not easy to miniaturize, but the invention of the transistor, whose behavior is based on quantum physics effects, made it possible to replace vacuum tubes and gave rise to a new generation of computers. Transistors are much smaller, require less power, last longer and rapidly replaced vacuum tubes to become the building blocks of computers and all modern electronics.

But what has quantum physics have to do with transistors? To answer this, we need to understand

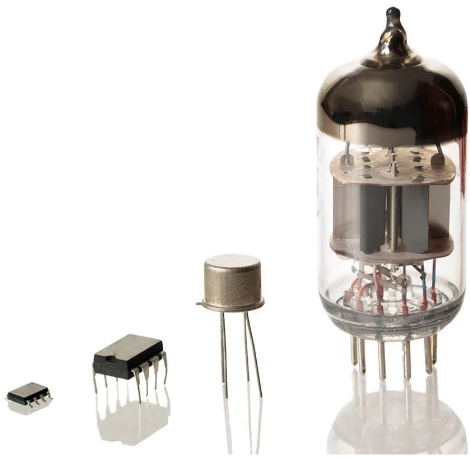

Figure 7.1: *A typical triode tube (on the right) has a size in the order of several centimeters (about 3-5 cm tall or more), and is often housed in glass. Modern transistors, especially those embedded in integrated circuits (on the left), can be on the nanometer scale internally, far smaller than any triode. Source: Krupczak Jr., Understanding Technological Systems.*

what transistors are made of. All transistors and in general all the components of modern microelectronics are made of materials called semiconductors (most of the time it's a silicon based material). Semiconductors, as the word says, are materials that have a property of being in the middle between a conductor, which allows the transmission of electricity, and an insulator, which completely blocks any electricity to flow through it.

We have seen how the electrons in atoms can only have certain discrete energy levels. We have seen it both as the result of Bohr atomic model and as a consequence of applying the Schrödinger's equation to a system made of a central positive charge orbited by a negative charge. When atoms get together to form molecules in solids, things get quite a lot more complicated as now the electrons of an atom are also feeling the attraction of nearby ones. What happens is that many more energy states become available and get very close to one another. So close that they form bands of energy. The band gap is an energy range where no electronic states are present, and thus no electrons can exist in this state.

In insulators, the valence band is separated from the conduction band by a large gap. This means that it takes a lot of energy (in other words a big voltage) to make electrons go from one band to the other and make conduction possible. In good conductors such as metals the valence band overlaps the conduction band, so even a small voltage makes electrons flow almost without resistance. In semiconductors there is a small gap between the valence and conduction bands, so small that electrons can easily jump from one band to the other one.

Without going into the details of how a semiconductor based device works, this narrow gap between the valence and conduction band allows controlling the flow of current in the material by applying a tiny voltage. This is exactly the principle of the transistor, a device where the current flow from one point to the other is controlled by the voltage applied to a third point. The fact that this voltage can be very small allows to operate these devices with very little energy and to make them as small as needed, at the level of a few atoms per device.

The first transistors' size was around 1 cm (or half an inch), which was between three and four times smaller than a typical vacuum tube. Today's transistors have a size of 1 nanometer, 4 million times smaller than the average vacuum tube. At these distances classical physics doesn't apply anymore and transistors start to show behaviors that can only be explained by quantum physics. The most important one is quantum tunneling.

Quantum tunnelling or simply tunnelling is a phenomenon predicted by quantum mechanics, where an object (for example an electron in a transistor) passes through an energy barrier even when it doesn't have enough energy to pass through it. In

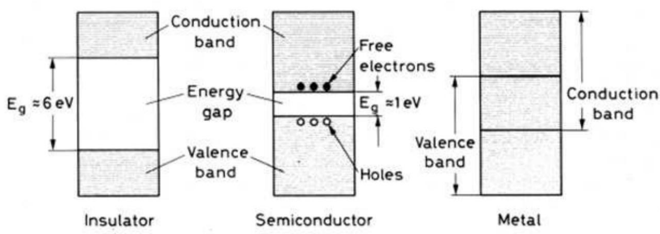

Figure 7.2: *From left to right, the band gap in insulators, semiconductors and conductors like metal. The energy difference between the valence band (where electrons are bound) and the conduction band (where electrons are free to move and conduct electricity) is large in insulators, quite small in semiconductors, and doesn't exist in metals, where the two bands overlap. Source: Tavernier, Experimental Techniques in Nuclear and Particle Physics.*

the case of a transistor, the barrier serves to control the current between two terminals. When this barrier extends over only a few atoms, the barrier is so thin that the probability of quantum tunneling becomes significant, causing electron leakage, by allowing a few electrons through even when the transistor is supposed to be off. This leads to inefficiencies in traditional silicon-based transistors and effectively limits how far traditional transistors can be scaled down without affecting their behavior.

This effect highlights the fact that it will be very difficult to continue to develop more powerful chips by simply increasing the miniaturization. Once quantum effects like tunneling become significant, no further miniaturization will be possible anymore, and new technology will have to be developed. Quantum computing is one of those.

7.2 Solar Cells

Solar energy is clean, very safe, reliable and pollution free. And if it wasn't for quantum physics, we would not be able to build a device like the solar cell, that is able to turn sunlight into energy, exploiting a mechanism called the photovoltaic effect.

The photovoltaic effect is a phenomenon very similar to the photoelectric effect that Einstein explained in terms of electrons absorbing a high frequency photon and being ejected from a metal. The photovoltaic effect instead usually happens in semiconductors like silicon. In this type of material, the absorption of light causes an electron excitation, but in this case the electron doesn't jump out, it just moves from the valence band to the conduction band. The electron that absorbs the photon energy is free to move around within the semiconductor, leaving a positive hole behind, its old place in the valence band. This place is rapidly occupied by another electron, leaving another positive hole behind it, and so on, generating an electric current inside the cell.

The electricity generated by solar cells can be stored inside a battery, which is in turn used as a source of power when needed. Solar cells are used as an alternative source of power in homes, but are also the main source of power for many spacecrafts, like the International Space Station.

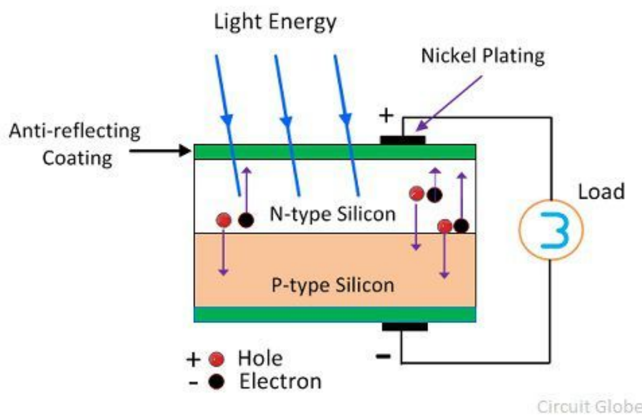

Figure 7.3: *Schematic view of a solar cell. When sunlight hits this cell, photons excite electrons in the silicon atoms, freeing them from the valence band and making them able to move inside the material, generating a current. Source: Circuit Globe*

7.3 LEDs

Semiconductors are also the base material for the production of LEDs, that is an acronym for Light Emitting Diodes. Those devices are based on the unique property of semiconductors, in which electron energy levels are essential split into two bands, the valence band at lower energies, where the electrons are tied to their atoms, and the conduction band, where the electrons are free to move inside the material.

The idea behind LEDs and diodes in general is to build a so-called p–n junction. In a p–n junction two types of materials are placed in contact with each other. The result creates a large band gap where the electrons from the n side migrate to the p-side. When an electric current flows through it the electrons from one side fall from the conduction to the valence band and in this process emit a photon, with a frequency corresponding to the width of the energy band gap. This process is called electroluminescence. Since the electrons jump from an energy band to another, and not from a specific atomic energy level, the energy of the photons emitted in the process are not always exactly the same, because the electrons can in principle come from any energy level available in the conduction band to any energy level available in the valence band. In practice, since the energy levels in the bands are quite close to each other, the frequency of the photon emitted is quite peaked around a specific color.

LEDs have been replacing incandescent light sources in everyday use. Their main advantage is lower power consumption. Incandescent light sources rely on heating a filament to a temperature

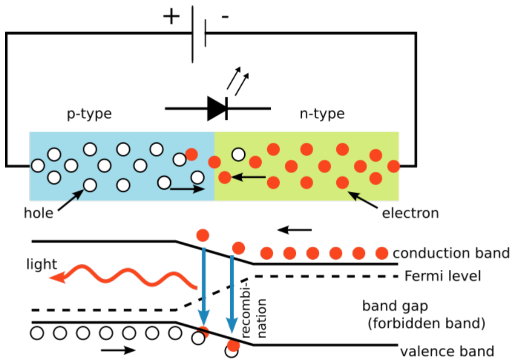

Figure 7.4: *LEDs (Light Emitting Diodes) work by a process called electroluminescence in a semiconductor p-n junction. Electrons from the n-type semiconductor recombine with holes in the p-type semiconductor at the junction. This recombination releases energy in the form of light. Source: S-key, Wikipedia.*

high enough to emit light in the visible spectrum (remember the black body radiation?). In the process, most of the energy needed to produce light is converted into heat, while in the case of LEDs the opposite is true. The electrons lose most of their energy by emitting a photon, and only a small part of the energy is converted into heat in the process. The advantages of LEDs also include a longer lifetime, higher robustness, and considerably smaller sizes than incandescent light sources.

7.4 Lasers

Lasers are devices that emit amplified light, by stimulating the emission of radiation from atoms. In fact, LASER is just an acronym for Light Amplification by Stimulated Emission of Radiation. The main property of the light coming from lasers is its coherence. The coherence of light waves means that these waves keep a fixed phase relationship as they propagate. Every photon in the laser beam is in the same narrow frequency range, and it has the same phase in the sense that the peaks and troughs of the light waves remain aligned as the waves propagate.

Lasers exists only because of quantum mechanics. We have seen that the quantization of energy in atoms was theorized by Bohr and then confirmed by various experiments. This means that electrons that are orbiting around the nucleus of an atom, can only have a finite set of energies and not a continuous spectrum. These electrons can absorb an emit energy, for example by absorbing a photon, but only if the energy carried by the photon matches the difference in the energy levels that are permitted. In general, these energy levels vary from an atom to another. Once a photon hits an electron, it jumps from an energy level to another, and the photon is completely absorbed in the process.

Now, what happens to the electron in this excited state? According to classical physics, there is no reason why the electron could not stay in this state forever. But quantum physics disagrees. According to quantum physics, eventually a photon will be spontaneously created with the exact energy to allow the electron to transition to a lower energy state. This emission process is called spontaneous emission. But there is another type of emission from an excited state. If a photon with the right frequency is shot at an electron that is already in an excited state, it can cause the electron to drop from the higher to the lower level. In this process, another photon will be emitted, and it will exactly match the original photon in frequency, phase, and direction. This is called stimulated emission, and it's responsible for the light emitted by lasers.

Lasers are built by creating a large population of atoms in excited states (atoms are in excited states if one of their electrons has absorbed a photon and jumped from an orbital to another with a higher energy). This can be achieved using an outside light source, or an electrical field. This medium made of excited atoms is placed inside an optical cavity, which is just a closed box with reflective mirrors, that doesn't let the light escape. Some of the excited electrons in the medium will then spontaneously emit photons, and some of the photons emitted will hit other atoms and stimulate photon emission. The photons emitted will have the same frequency, direction and phase of the original one and will be reflected by the mirrors in the cavity. This will create a cascade effect, where more and more photons with the same properties

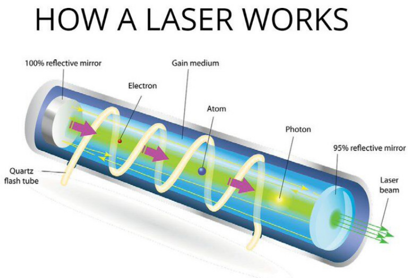

Figure 7.5: *A simplified laser. The gain medium is inside a cylinder with reflective mirrors. A quartz tube is wrapped around it and photons emitted from it excite electrons in the medium. These electrons will then transition to a lower energy state and emit photons, which in turn will stimulate the transition of other electrons in the medium. This creates a cascade effect that amplifies the number of photons in the cavity. The laser beam is extracted by creating a narrow hole on one side of the cylinder. Source: Designua.*

will be emitted and in turn stimulate the emission of other identical photons. But how to extract these photons from the cavity? The final trick is to make one mirror fully reflective, and the other partially so it lets a portion of the light escape. This light is what the laser beam is made of.

Today, lasers are everywhere. They are used in medicine, like in corrective eye surgery or tattoo removal. But they are also widespread in industrial assembly lines, supermarket scanners, optical communications, and optical data storage.

7.5 Quantum Computing

We mentioned earlier that the development of more and more powerful computers has been achieved essentially by increasing the miniaturization of components and fitting more and more transistors in a single chip. In fact, this trend has been going on for such a long time that there is a law that describes it. Moore's law states that the number of transistors in an integrated circuit doubles about every two years. Of course this is not a physical law, but more an empirical observation, Nevertheless, this prediction has held since 1975 and has even been used in the semiconductors' industry to define long term plans and set research targets. Quantum physics might soon put an end to it.

As the miniaturization reaches the nanometer scale (which is a billion times smaller than a meter) effects like quantum tunnelling start to appear. As we have seen, because of the quantum tunneling, electrons that are not supposed to cross a potential barrier, for example in a transistor that is supposed to be off, nevertheless cross this barrier and cause electron leakage. This can make the output of a logic port flip from zero (no current) to one (some current), leading to computational errors and unpredictable behavior in the components of a traditional microchip. What will happen once we reach this threshold? Most probably this will accelerate the development of quantum computing even further, and we are seeing signs of this already today.

A quantum computer operates on quantum mechanics principles. In particular, the unit of information in quantum computing, the qubit, can exist in a state which is a superposition of traditional bit states (the classical 0 and 1). For example, a qubit can exist in a state that is 50% zero and 50% one. The advantage of a quantum computer is that it's able to operate on all bit states at the same time. For example, let's say we want to find the result of applying a function to a number that can be represented by a single qubit. In the case of a classical computer, that operates on traditional bits, we have to apply the function two times to know the result, once when the bit value is 0 and another when the value is 1. Quantum computers operate with qubits, that are a superposition of traditional bits. So the function will be applied to all possible states at once. This dramatically reduces the time needed to perform a calculation and the gain in performance is higher as the complexity of the calculation increases and the number of bits involved grows.

The result of the computing process is finally extracted by performing a measurement, which makes the wave function of the qubit collapse to a single state (either zero or one for each qubit). As we have seen, this process is not deterministic, and the result of the measurement will return some time the qubit in a zero state and some other time in a one state. The key to quantum computing is to exploit this feature by manipulating the qubits according to various algorithms, with the aim of amplifying the probability of the desired measurement result. Over the last decades, quantum algorithms have been developed to solve various tasks, like breaking encryption protocols, or speeding up search problems, or even simulate quantum systems faster than normal computers.

In the last years quantum computing has moved fast from more theoretical development towards more practical, deployable machines capable of sustained operation and more effective error correction. Recently Harvard and MIT researchers developed the first-ever quantum computer that can operate continuously without restarting. The machine is made of 3000 qubits and can replenish lost qubits by injecting fresh atoms, overcoming the problem of qubit loss. This and other similar achievements mark a fundamental step forward in

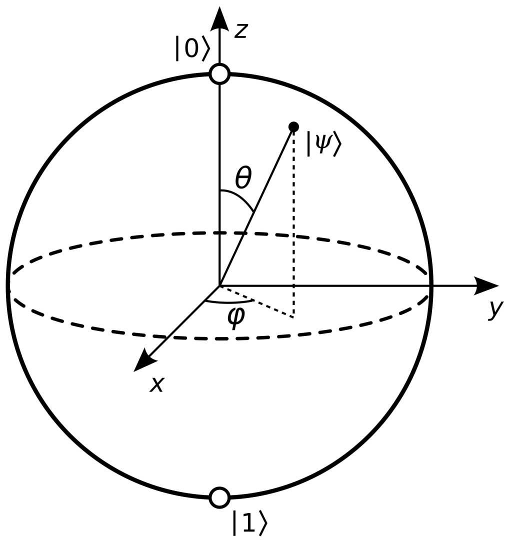

Figure 7.6: *The Bloch qubit representation uses the Bloch sphere, a geometrical visualization tool for the state of a single qubit in quantum computing. The Bloch sphere is a unit sphere where every point on its surface corresponds to a pure qubit state, represented as a combination of the two basis states 0 and 1. Source: Wikipedia.*

the stability and usability of quantum computers, moving them closer to practical, long-term use.

7.6 Everyday Quantum Physics: 4 takeaways

> The technologies used in the early experiments related to quantum physics have been greatly developed in the last 100 years or so, and are now part of our everyday life.
>
> Semiconductors are materials were electrons can exist in two energy bands, the conduction band where the electron is free to move, and the valence band where the electron is tied to an atom, making it a conductor and an insulator at the same time. This allowed us to build devices like the transistor, where the flow of current can be manipulated by applying small voltages. Transistors are the base of logic ports that constitute the building blocks of all modern electronic devices

Semiconductors make also possible to build devices like the LEDs, that produce light by forcing the flow of electrons from a high energy band to another and in the process emit a photon. The inverse process, the absorption of a photon by an electron, and its jump from the valence band to the conduction band, makes it possible to build devices like solar cells, that transform sunlight into electrical current and ultimately produce energy.

Lasers rely on amplified radiation emission from atoms. The basic functioning principle is to cause electrons in atoms to jump from a low energy to a higher energy level. Once the electrons fall back to the lower energy level, they emit photons and this photon in turn can hit an excited electron from another atom and cause the emission of an identical photon in the same direction, which will in turn go and stimulate another photon emission. This cascade process creates a high number of photons with the same frequency, confined in a narrow space, which makes up the typical laser beam.

Quantum computers rely on quantum physics. Instead of bits, they operate on qubits, that can exist in a state which is a superposition of traditional bit states. The key to quantum computing is to exploit this superposition by manipulating the qubits, with the aim of amplifying the probability of the desired measurement result.

7.7 To know more

- If you are not scared of a bit of math, or you just want to know how do you get from the Schrödinger's equation to the electronic band structure in materials, this extract from a book by Cardona
  - https://tinyurl.com/ycrmnfej
- This is another fascinating historical article on the discovery of LEDs. The first observations of light emission from a semiconductor date back to the 1920s.
  - https://tinyurl.com/3dz3u69u
- The history of the invention of the transistor and where it will lead us. It has a lot of good details on how scientists and engineers were able to overcome various issues in the

transistor miniaturization process in creative ways.

  - https://tinyurl.com/3d62563f

- This is probably the best written and more detailed article on the photovoltaic effect, the principle behind solar cells.
  - https://www.chemistryexplained.com/Ru-Sp/Solar-Cells.html
- This is a bit of an advanced paper, by Huang & Mittal. It discusses the effects of quantum tunneling in the miniaturization of transistors. It can get technical in some points, but I believe it's still suitable for the general audience.
  - https://tinyurl.com/mrpwyf4j
- This free book by Thomas Wong is an excellent introduction to classical and quantum computing. This book is suitable as a course textbook or for independent study.
  - https://www.thomaswong.net/introduction-to-classical-and-quantum-computing-1e4p.pdf
- This video is worth watching if you want a detailed and exhaustive explanation of how a quantum computing algorithm works.
  - https://youtu.be/RQWpF2Gb-gU?si=CS6tyHZeVhCm4Hft

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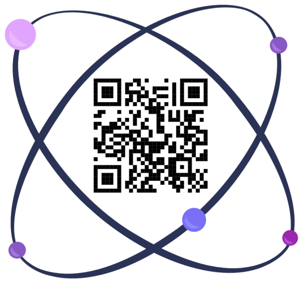

8 Interpretations of Quantum Physics

Quantum Physics is a successful theory, it can explain virtually all observed phenomena, but it doesn't really give a clear indication about the nature of the physical world that it describes. In this sense, it is maybe unique among physical theories, as it still needs an interpretation of what it actually means in relation to the world it describes. The interpretation of quantum physics kept the best Physicists of the early 20th century busy. Bohr, Einstein, Heisenberg and others devoted many energies and discussions in the attempt to understand if the principles of quantum physics corresponded to the nature of reality or were just a consequence of our limited understanding of reality at the microscopic level. During those early times this seemed a very important problem, as important as the mathematical formulation of the theory itself, but the reality is that, once this generation of physicists passed and was replaced by a new one, more accustomed to the new ideas, the debates about the true nature of reality gradually faded away.

I also want to bring my personal experience here. As a Physics student, these problems were definitely never discussed during the lectures. They were maybe discussed among students, but, even then, they were considered philosophical questions, not worth much attention. The so-called "shut-up and calculate" approach, advocated by Feynman among others, was definitely the most popular (and easiest) way to solve the question. The idea was to just shut up and focus on the mathematical aspects of the theory, trying to solve real problems and hoping that they could lead to useful insights and predictions. Despite having an immense admiration for Feynman, I don't agree with this approach, and I think that it's important to reach an agreement on a unified interpretation of quantum physics, not only because it will make it more easily understandable, but also because it might lead to deep insights on the nature of reality, like the theory of relativity changed our understanding of the nature of space and time.

In this chapter I am going to go through the most popular interpretations in detail. Some of them, like the Copenhagen interpretation and the many-worlds one, are based on the standard quantum physics theory developed by Heisenberg and Schrödinger. Others, like the hidden variable theories, propose a slight modification of the standard theory, so they represent not only a different interpretation, but a different formulation of Quantum Physics.

8.1 The Copenhagen Interpretation

The earliest interpretation of quantum physics emerged from the ideas of Bohr and Heisenberg in Copenhagen during the 1920s, thus it became known as the Copenhagen interpretation. Bohr was of the idea that our understanding of the world is necessarily classical, as we perceive the world as consisting of objects (such as waves or particles) moving in a three-dimensional space, and this is the only framework available to us and accessible to our senses. This framework completely fails for Quantum physics, which does not allow for any description of matter in classical terms, whether waves or particles. In conclusion, according to Bohr, the quantum realm is fundamentally beyond our direct comprehension. Quantum physics should not be seen as an accurate depiction of the quantum world, nor should the time evolution of the quantum state be interpreted as a causal account of observed phenomena. Instead, quantum physics should simply be seen as a highly successful predictive tool that takes the classical description of measurement apparatus as input and outputs probabilities of possible classical measurement outcomes.

Probably the Copenhagen interpretation was influenced by the logical positivism popular in Europe during the same time. Logical positivists claimed that the meaningful content of a scientific theory lies solely in its empirical predictions, and any speculation about the nature of reality is meaningless. However, Bohr interpretation is a bit more nuanced. He definitely accepted that a micro-world governed by quantum physics laws existed, and he also acknowledged that it was not understandable in terms of deterministic cause-effect relations. Like Kant, Bohr thought that we as humans are limited to certain conceptual frameworks, and the world as it truly is, escapes these frameworks. If so, then the inability of our fundamental physical theories to faithfully describe reality is expected, and our failure to interpret quantum physics as a literal description should not trouble us.

The Copenhagen-style interpretation can be justified without committing to any philosophical doctrine. For example, it's widely agreed on, that the wave function's intensity at a point gives the probability of finding a particle there. It is then natural to consider the wave function as representing our knowledge of the system rather than the system itself. This view (which was also supported by Einstein) implies that quantum physics is incomplete. In other words, it's just a tool that allows us to calculate probabilities, not a description of the underlying state that produces these probabilities. Later findings showed that under plausible assumptions, describing such an underlying state is impossible. This difficulty suggests that the quantum world might remain impenetrable to us.

However popular, the Copenhagen interpretation faces an important challenge. Bohr's view clearly defines a border between the classical macroscopic world, governed by determinism, and the microscopic quantum realm, in which the deterministic view completely fails. But of course the macroscopic world must arise from the microscopic, in some way. Since macroscopic objects consist of microscopic parts, it seems macroscopic objects must also obey quantum physics, so no clear dividing line can exist delineating where quantum rules apply and where they do not.

8.2 The Many-Worlds Interpretation

Hugh Everett, in 1957, introduced a groundbreaking interpretation of quantum physics that treats the quantum state as a complete and real description of physical systems, including macroscopic ones, eliminating the need for any frontier between micro and macro levels. This radical approach proposes that all possible outcomes of quantum measurements actually occur across different branches of a continually splitting multiverse, known as the many-worlds interpretation. According to Everett, the collapse of the wave function is not real. Instead, what happens is that the universe's state evolves linearly, resulting in multiple, non-interacting branches, each representing a different outcome.

To solve the apparent contradiction of this model, which is the fact that the outcome of a measurement always leads to a definite result, Everett suggested modeling the measurement process within the quantum framework itself. In other words, he suggests including devices and observers in the measurement process. For instance, if the wave function intensity of an electron at the end of an experiment is non-zero only in two regions, A and B, then the detectors and observers are also represented as wave functions. Interaction causes a superposition of outcomes, where the system, detectors, and observers become entangled in a single wave function. This idea implies that the entire system, including human observers, exists in a superposition of distinct states, corresponding to different measurement outcomes, with each "branch" evolving independently and not influencing each other.

This interpretation overcomes the traditional problem of having two distinct worlds, the quantum and classical. In fact, it considers the entire universe as subject to quantum physics without a collapse, but instead branching out at each observation. In this universe, all outcomes occur, but observers perceive only one because they are part of a specific branch. Of course, the many worlds interpretation raises huge philosophical questions about personal identity and probability. For instance, if a person splits into two copies, the issue arises whether they are the same individual or different, with solutions varying from denying strict identity to viewing persons as extended in four-dimensional histories or as branching entities.

A major challenge involves understanding how probabilities function if all outcomes occur—each with a "branch" in the multiverse. Quantum physics predicts probabilities based on the wave function's squared amplitude, but in a universe where every outcome is realized, the meaning of probability becomes obscure. Some Physicists and Philosophers, like David Deutsch and David Wallace, have argued that rational agents should act as if these squared amplitudes are genuine chances, but this view faces criticism regarding the rationality of caring about branches.

It can be argued that the many-worlds interpretation is extravagant to say the least, because it postulates countless copies of objects and individuals. However, the prevailing view nowadays is that worlds are emergent features of the quantum state itself, not separate entities. Wallace also emphasizes that the number of branches is not well-defined and that this indeterminacy can mitigate concerns about the interpretation's ontological commitments, framing worlds as features rather than distinct entities.

This interpretation remains controversial but offers a radical, realist view of quantum physics where the universe's structure embodies the full richness of all possible outcomes, providing a coherent, if philosophically challenging, resolution to the measurement problem.

8.3 Hidden Variable Theories

The many-worlds interpretation suggests that believing a quantum measurement has a single definite outcome is mistaken. Instead, such measurements produce multiple outcomes across different branches of reality. However, this concept might be difficult to accept, especially considering unresolved issues related to identity and probability. Consequently, some conclude that quantum mechanics is incomplete because nothing in the quantum state singles out one particular measurement outcome as the definitive one. This perspective, historically associated with Einstein, implies the need to supplement quantum mechanics with additional variables, often called hidden variables, which describe the actual state of the world.

Hidden variable theories were quite popular up to the 40s and 50s, but their popularity received quite a blow when, in 1964, John Bell came up with a theorem demonstrating that, under reasonable assumptions, no hidden-variable theory can fully complete quantum mechanics. His proof involves properties such as spin of pairs of particles, showing the impossibility of assigning predetermined values to measurements that would reproduce quantum predictions. While many physicists interpreted

Bell's theorem as ruling out hidden variables, Bell himself concluded that one of his assumptions, for example locality or measurement independence, could be false. The locality assumption means that measurements on one particle cannot instantly influence another distant particle, which seems safe to assume, as such a connection in principle implies an instant transfer of information between the two particles. The assumption of measurement independence implies that particle properties are independent of which measurements are chosen, which also seems as a safe assumption, given that the choice of measurement can be made using a randomizing device for example. For these reasons, any hidden variable theory would have to violate one of these assumptions, making it very difficult to accept.

An example of a hidden variable theory is the one developed by David Bohm in 1952. Bohm's model effectively explains measurement outcomes and interference phenomena by assuming that real particle positions are guided by a quantum wave. It circumvents Bell's theorem by violating locality and introducing a dynamic where the position of each particle depends on all others instantly. This conflicts with special relativity's constraints, because it implies the existence of an instantaneous correlation between particles. One way to solve it is to modify special relativity itself and introducing the concept of simultaneity but, again, this creates more problems than it solves.

Some hidden variable theories extend beyond particle positions (used by Bohm) to include other properties, but it has been proven that no theory assigning definite values to all properties simultaneously can reproduce quantum mechanical results. Modal theories attempt to resolve this by dynamically selecting which properties have definite values during measurement without providing explicit dynamical laws. Other approaches, like retrocausal theories, reject the independence assumption by allowing future events to influence past properties, helping reconcile hidden variables with relativity without instantaneous action at a distance. These theories propose that causal influences from both past and future jointly produce quantum effects, although they remain speculative without fully formulated dynamics.

8.4 Spontaneous Collapse Theories

Spontaneous-collapse theories take the wave function as a complete description of a system but modify the Schrödinger dynamics so that superpositions are occasionally and spontaneously reduced to localized states. The motivation is simple, the Schrödinger equation is linear, so superposed inputs produce superposed outputs, which makes measurement outcomes multiplicative unless the dynamics itself breaks linearity. Spontaneous-collapse models introduce non-linear terms that are responsible for suppressing macroscopic superpositions while leaving microscopic quantum behavior essentially intact.

The prototypical model is the GRW (Ghirardi–Rimini–Weber) theory (developed in 1986). In GRW each particle has a tiny probability per unit time to undergo a sudden, random localization. Mathematically, the wave function is multiplied by a narrow function centered at a randomly chosen position (the collapse center). The collapse rate per particle is vanishingly small (of the order of one collapse every 100 million years for a single particle) so microscopic interference survives. But collapse events scale with particle number, so macroscopic objects that have billions and billions of particles undergo extremely frequent effective localizations, rapidly producing well localized, effectively classical states.

This mechanism explains why interference is seen in microscopic experiments while measurements yield single, definite outcomes. Individual quantum systems evolve nearly unmodified by GRW until they become entangled with many degrees of freedom, at which point the spontaneous localization rapidly selects a localized branch. Unlike hidden-variable approaches, GRW contains no additional particle ontology by default—the apparent particle-like outcomes emerge from the wave function's localized profile after collapse.

GRW and related collapse proposals face several important challenges:

- The "tails" problem: localization never strictly annihilates non-chosen branches. Small amplitude components remain nonzero. Interpreting these tails raises questions about whether outcomes are truly unique and how probabilities should be understood.
- Ontology: the universal wave function lives in a high-dimensional configuration space, which prompts concerns about how the familiar three-dimensional world emerges. One response is to posit an additional three-dimensional mass-density field determined by the wave function so that empirical objects are directly represented in physical space.
- Non-locality and relativity: collapses act effectively instantaneously across space, violating Bell's locality assumption and introducing compatibility problems with special relativity. Relativistic generalizations of collapse dynamics are an active area of research but remain challenging.

Collapse models also make experimental predictions that differ subtly from standard quantum mechanics. Because collapse events introduce small deviations from unitary evolution (for example, tiny energy increases or spontaneous heating), precise interferometry, matter-wave experiments, and optomechanical tests can constrain the allowed collapse parameters. Ongoing and planned experiments continue to tighten these bounds.

8.5 Other interpretations

Several interpretations do not fit neatly into the earlier categories. Below are concise summaries of the main alternatives, their core ideas, and key strengths or challenges.

The Consistent (Decoherent) Histories (Griffiths, Gell-Mann, Hartle, Omnès) interpretation assigns probabilities to entire sequences of events (called histories) provided interference between those histories is negligible (consistency/decoherence condition). For systems strongly coupled to an environment, decoherence suppresses interference and histories can receive ordinary probabilities. The approach permits many different, mutually incompatible sets of histories. Proponents treat one consistent set as describing the actual course of events, while critics ask how a unique description of reality emerges and whether plural descriptions can be coherent.

The Transactional Interpretation, originally proposed by John Cramer, uses both forward-in-time and backward-in-time wave solutions. Transactions form when retarded (forward) and advanced (backward) waves establish a handshake between emitters and absorbers. One completed transaction is taken as the actual event. The retrocausal element can evade Bell-type constraints, but questions remain about how and why a single transaction is realized and whether the picture provides a satisfying dynamical account of measurement.

The Relational Interpretations, developed by Carlo Rovelli and others (and promoted in different forms by Mermin) hold that quantum properties are not absolute but only make sense relative to other systems. There are no observer-independent values, because a measurement outcome is a relation between system and observer. This reframes non-locality and avoids assigning global definite properties, but it challenges traditional notions of a single, objective world and raises questions about intersubjective agreement and how classical objectivity emerges.

The Informational Interpretations, supported by authors like Bub, Caves, Fuchs and Schack, treat quantum theory as primarily about information, probabilities, or agents' beliefs rather than about intrinsic properties of isolated systems. Variants range from instrumentalist readings (quantum theory as a rulebook for making predictions) to realist information-first accounts that elevate "information" to a primitive. The main virtue is conceptual clarity about what the formalism constrains. The main criticism is that an informational stance may omit or postpone a physical account of how a single, objective world evolves.

Each of these approaches trades off different metaphysical commitments (realism, locality, temporality, observer-independence). None has achieved universal acceptance. Together they illustrate the range of conceptual strategies for addressing measurement, probability, and the emergence of classicality in quantum mechanics.

8.6 Quantum Physics Interpretations: 4 takeaways

The Copenhagen interpretation is the earliest interpretation of quantum physics. Bohr was of the idea that our understanding of the world is necessarily classical, and the quantum realm is fundamentally beyond our direct comprehension. Quantum physics should simply be seen as a highly successful predictive tool that takes the classical description of measurement apparatus as input and outputs probabilities of possible classical measurement outcomes.

Everett introduced the Many-Worlds interpretation in 1957. According to this approach, the quantum state is a complete and real description of physical systems, including macroscopic ones. This implies that all possible outcomes of quantum measurements actually occur across different branches of a continually splitting multiverse. The collapse of the wave function is not a real phenomenon, and what really happens after a measurement, is that the universe splits in multiple, non-interacting branches, each representing a different outcome.

Hidden variable theories start from the assumption that quantum mechanics is incomplete, and we need to supplement it with additional variables, often called hidden variables, which describe the actual state of the world. However, in 1964, John Bell demonstrated that, under reasonable assumptions, no hidden-variable theory can fully complete quantum mechanics. Bell himself didn't consider this a definitive blow to hidden variable theories, pointing out that maybe one of his assumptions, for example locality or measurement independence, could be false.

Spontaneous-collapse theories take the wave function as a complete description of a system but modify the Schrödinger dynamics so that superpositions are occasionally and spontaneously reduced to localized states. Spontaneous-collapse models introduce non-linear terms that are responsible for suppressing macroscopic superpositions while leaving microscopic quantum behavior essentially intact.

Many more interpretations have merged in recent years. Each of these approaches trades off different metaphysical commitments (realism, locality, temporality, observer-independence), but none has achieved universal acceptance. Together they illustrate the range of conceptual strategies for addressing measurement, probability, and the emergence of classicality in quantum mechanics.

8.7 To know more

- A very deep discussion of the Many-Worlds interpretation of Quantum Physics can be found in this article,
  - https://plato.stanford.edu/entries/qm-everett/
- All those interpretations (and more) are summarized in this excellent article.
  - https://iep.utm.edu/int-qm
- A great overview of Bohm's alternative interpretation of Quantum Physics.
  - https://plato.stanford.edu/entries/qm-bohm/

- One of the loudest advocates of the Many-Worlds Interpretation in the recent time is Sean Carroll. You can find a lot of very good material on the subject in his blog.
  - https://www.preposterousuniverse.com/blog

Scan the QR code below to access all additional content

9 The learning path ahead

When I first approached Quantum Physics, way before I started to study Physics seriously, it seemed to me an enigmatic and complex subject, completely removed from everyday experience. But I hope that, throughout this book, I managed to show you that Quantum Physics is fundamentally about reality at its most basic level, it's about the behavior of the tiny particles that form everything that surrounds us. In this conclusion I want to sum up the core insights and reflect on their significance, plus giving you a few pointers to satisfy your curiosity and keep you on this amazing learning path.

Classical physics gave us an intuitive, predictable map of nature's workings, from the motion of planets to the behavior of mechanical devices. However, the dawn of the 20th century brought revolutionary experiments and ideas that shook the foundations of that classical worldview. Quantum physics emerged as a new framework to describe phenomena that classical physics couldn't explain: discrete energy levels, wave-particle duality, uncertainty, entanglement, and more.

I have shown you how quantum particles behave both like waves and particles, how electrons inhabit discrete energy states in atoms, and how measurement influences the system being observed. These ideas challenged traditional logic and forced scientists to rethink the nature of reality itself.

9.1 The key takeaways

While the mathematics behind quantum physics can be demanding, the idea that at the smallest scales nature is probabilistic and interconnected has profoundly altered how we understand the universe. Just let me remind you the key concepts that make Quantum Physics fundamentally different from our macroscopic world.

1. **Wave-Particle Duality:** Particles such as electrons and photons exhibit both wave-like and particle-like properties, depending on how we observe them. This duality lies at the heart of quantum mechanics and unites two classical concepts that seemed incompatible with each other.

2. **Quantum Superposition:** Quantum systems can exist in multiple states simultaneously until measured. Schrödinger's cat

paradox illustrated this strange but fundamental idea that quantum states persist as a combination of possibilities.

3. **Uncertainty Principle:** Heisenberg's uncertainty principle states that you cannot simultaneously know certain pairs of properties (e.g., position and momentum) with perfect accuracy. This limitation is not due to measurement flaws but a fundamental aspect of reality.

4. **Quantum Entanglement:** Particles can become entangled, so that the state of one immediately influences the state of the other, no matter how far apart they are. This phenomenon still challenges physicists nowadays, as it suggests that laws of nature might be non local.

5. **Quantization of Energy:** Energy in bound states like atoms, exists in discrete units or quanta, such as photons of light, explaining atomic spectra and leading to technologies like lasers.

6. **Probability and Measurement:** Outcomes in quantum experiments are inherently probabilistic, described by a wave function that encodes the likelihood of possible results. Measurement causes the wave function to collapse to a specific outcome.

9.2 How to go forward

Quantum physics is not just a theoretical curiosity, it underpins much of modern technology. Without it, inventions like semiconductor devices, computers, lasers, MRI scanners, and even GPS systems could not exist. Exploring quantum physics reveals how the tiny building blocks of the universe give rise to the complex macro world we inhabit.

More so, quantum physics challenges our philosophical understanding of reality. The idea that observation affects outcomes invites questions about the role of consciousness, the nature of causality, and the limits of human knowledge. Even if the Academic Physics community has preferred not to acknowledge this fact for a long time, the debate around these topics is still ongoing to this day.

Although this book introduced the basics, Quantum Physics is a vast and evolving field. For those intrigued by what you've learned, many paths are open for deeper study. Here I list the main ones, together with a few reading suggestions.

- **Mathematical Foundations:** Delving into linear algebra, differential equations, and complex numbers will allow you to understand quantum mechanics in its full mathematical beauty. If you are interested in this aspect of the theory, you will love books like "Calculus Made Easy" by

Thompson, and “Linear Algebra done right” by Sheldon Axler.

- **Quantum Computing:** Learning about the hardware behind the next generation of computers and the challenging algorithmic problems will give you an insight into the power of a technology deeply connected to quantum mechanics and the principle of superposition of states. A good introductory book on the subject is “Quantum Computing for Everyone” by Chris Bernhardt.
- **Interpretations of Quantum Mechanics:** Explore different philosophical viewpoints—such as the Copenhagen interpretation, many-worlds theory, and pilot-wave theory—that attempt to explain the weirdness of quantum effects. “Physics and Philosophy” by Werner Heisenberg is the classical book to read if you are interested in understanding the implications of Quantum Physics on our view of the world.
- **Quantum Field Theory:** This is the framework that merges Quantum Physics with special relativity, explaining particles as excitations in fields. A very good introductory book is Sean Carroll’s “Quanta and Fields”.

The questions posed by Quantum Physics have led to profound technological advances and fresh insights about our place in the cosmos. They encourage us to keep questioning, keep experimenting, and keep imagining. As a beginner on this journey, you have taken important first steps into understanding the universe's fundamental workings. Whether your interest turns professional or remains a lifelong fascination, Quantum Physics offers a captivating window into nature's deepest secrets.

If this book inspired even just one of my readers, I will consider it successful. I hope this book inspired you too, and maybe pushed you a bit in the direction of pursuing your curiosity further, and deepened your understanding of the subject. As you keep studying, you will maybe struggle at times and find some difficulties along the path. My advice is not to worry and to enjoy the journey. As you have read, even the best minds have had issues understanding some aspects of the Quantum world. But perseverance and motivation can really go a long way, as I myself have experienced throughout my career as a PhD and Researcher in Particle Physics. Stay curious, keep learning, and more importantly, enjoy the wonder of the quantum world.

Thank You!

If you have made it to this page, let me Thank you for purchasing my book from the bottom of my heart.

I know there are dozens of books about Quantum Physics out there, but you gave mine a chance, you read it till the end, and I am very grateful for this.

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